

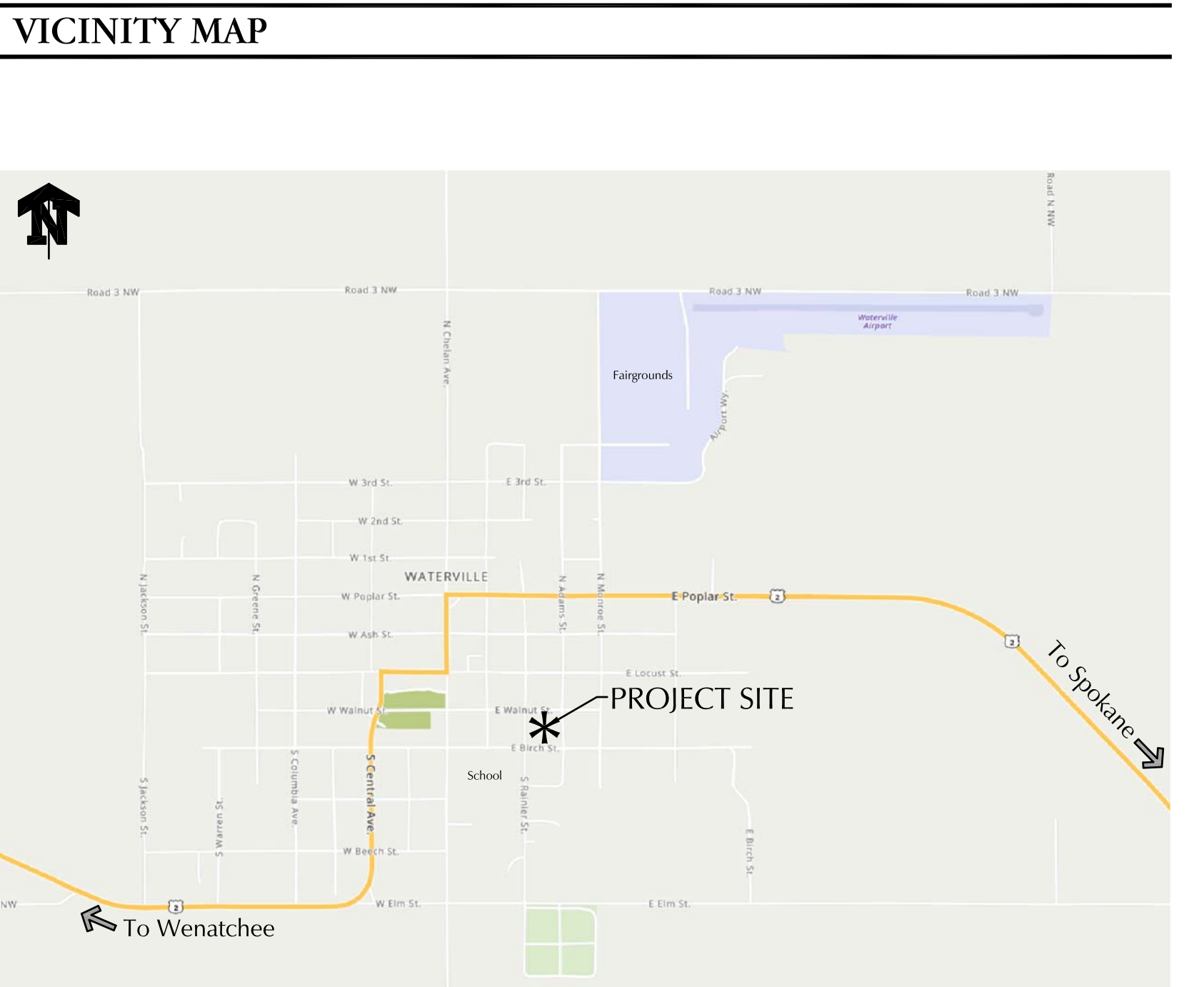
ABBREVIATIONS			
∕	Angle	JAN.	Janitor
Centerline	JT.	Joint	
Diameter or Round	LAM.	Laminate	
Square	LAV.	Lavatory	
Existing	LT.	Light	
Property Line	MAS.	Masonry	
A.B.	Anchor Bolt	MATL.	Material
A.C.P.	Asphaltic Conc. Paving	M.B.	Marker Board
A.D.	Area Drain	MAX.	Maximum
ADJ.	Adjustable	MECH.	Mechanical
A.F.F.	Above Finish Floor	MFR.	Manufacturer
AL.	Aluminum	MH.	Manhole
ALT.	Alternate	MIN.	Minimum
APPROX.	Approximate	MISC.	Miscellaneous
ARCH.	Architectural	M.O.	Masonry Opening
A.T.	Acoustical Tile	M.R.	Mirror
A.W.P.	Acoustical Wall Panel	MTD.	Mounted
BD.	Board	MTL.	Metal
BLDG.	Building	N.	North
BLKG.	Blocking	N.I.C.	Not in Contract
B.M.	Bench Mark	NOM.	Nominal
CAB.	Cabinet	N.T.S.	Not to Scale
C.A.B.	Cement Asbestos Board	O	Operable
C.B.	Catch Basin	O.C.	On Center
C.G.	Corner Guard	O.D.	Outside Dia.
C.H.	Coat Hook	O.F.C.I.	Owner Furn/Contractor Installed
C.I.	Curb Inlet	O.F.O.I.	Owner Furn/Owner Installed
C.I.P.	Cast in Place	OPNG.	Opening
C.J.	Control Joint	OPP.	Opposite
CLF.	Chain link Fence	O.R.D.	Overflow Roof Drain
CLG.	Ceiling	P.B.	Particle Board
CLR.	Clear	PC.	Precast
CMU	Conc. Masonry Unit	PIP.	Poured in Place
C.O.	Clean out	PL.	Plate
COL.	Column	P. LAM.	Plastic Laminate
CONC.	Concrete	PLAS.	Plaster
CONN.	Connection	PLYWD.	Plywood
CONST.	Construct(ion)	PNT.	Paint
CONT.	Continuous	P & P	Patch & Paint
DBL.	Double	PR.	Pair
DCD	Diper Changing Deck	P.T.	Pressure Treated
D.F.	Drinking Fountain	P.T.D.	Paper Towel Disp.
DIA.	Diameter	R.	Rise(r)
DIM.	Dimension	R.D.	Roof Drain
DISP.	Dispenser	REF.	Reference
DN.	Down	REFER.	Refrigerator
DR.	Door	REINF.	Reinforced
D.S.	Down spout	REQD.	Required
DTL.	Detail	RESIL.	Resilient
DW	Drywell	RM.	Room
DWG.	Drawing	R.O.	Rough Opening
DWR.	Drawer	R/S	Rod & Shelf
E.	East	R.W.L.	Rain Water Leader
EA.	Each	S.	South
E.B.	Expansion Bolt	S.A.P.	Solid Acrylic Polymer
E.F.	Exhaust Fan	S.B.	Smart Board
E.H.	Exhaust Hood	S.C.	Solid Core
E.J.	Expansion/Construction Joint	SCHED.	Schedule
E.S.	Emergency Shower	S.D.	Soap Dispenser
ELECT.	Electrical	SECT.	Section
ELEV.	Elevation	SHT.	Sheet
EQ.	Equal	S.J.	Slab Joint
EQUIP.	Equipment	SIM.	Similar
E.R.D.	Existing Roof Drain	S.N.D.	Sanitary Napkin Dispenser
EXIST.	Existing	S.N.R.	Sanitary Napkin Receptacle
EXP.	Expansion	SPEC.	Specification
EXT.	Exterior	SQ.	Square
F.A.	Fire Alarm	STD.	Standard
F.D.	Fl oor Drain	STL.	Steel
F.F.	Finish Floor	STOR.	Storage
F.E.	Fire Extinguisher	STRCT.	Structural
F.E.C.	Fire Exting.Cab	SUSP.	Suspended
FIN.	Finish	SYM.	Symmetrical
FLASH.	Flashing	S.S.	Stainless Steel
FLR.	Floor	T.	Tread
FLUOR.	Fluorescent	T.B.	Tack Board
FND.	Foundation	T.C.	Top of Curb
F.O.C.	Face of Conc.	T.D.	Towel Dispenser
F.O.F.	Face of Finish	TEL.	Telephone
F.O.S.	Face of Studs	T. & G.	Tongue & Groove
F.R.	Floor Register	T.I.	Tenant Improvement
FRL.	Fiberglass Reinf. Laminate	T.O.P.	Top of Parapet
FRP.	Fiberglass Reinf. Panels	T.O.C.	Top of Concrete
FT.	Foot or Feet	T.P.	Toilet Paper Dispenser
FTG.	Footing	TSC	Toilet Seat Cover Disp.
FURR.	Furring	TYP.	Typical
FUT.	Future	U.	Urinal
FWC	Fabric Wall covering	U.O.N.	Unless Otherwise Noted
GA.	Gauge	V.	Vent
GALV.	Galvanized	V.B.	Vapor Barrier
G.B.	Grab Bar	V.C.T.	Vinyl Composite Tile
GND.	Ground	VERT.	Vertical
G.W.B.	Gypsum Wallboard	VIF	Verify in Field
HB.	Hose Bibb	VTR	Vent Through Roof
HCP.	Handicapped	VWC	Vinyl Wall covering
H.C.	Hollow Core	W.	Waste Vent
HDW.	Hardware	W/	With
HDWD.	Hardwood	WB	White board
HGT.	Height	W.C.	Water Closet
H.M.	Hollow Metal	W.D.	Wood
HORIZ.	Horizontal	W.H.	Water Heater
H.S.	Heat Strengthen	W/O	Without
I.D.	Inside Diameter	WP.	Waterproof(ing)
I.H.	Intake Hood	W.R.	Water Resistant
INSUL.	Insulation	WT.	Window Treatment
INT.	Interior	W.W.F.	Welded Wire Fabric
		X	Fixed

ARCHITECTURAL SYMBOLS	
	PROPERTY LINE / CORNER
	EXISTING POINT ELEVATION (To Remain)
	FINISH POINT ELEVATION
	DRAINAGE COURSE (ON GRADE)
	NATURAL GAS
	WATER
	UNDERGROUND ELECTRICAL
	OVERHEAD ELECTRICAL/COMMUNICATIONS
	SANITARY SEWER
	STORM DRAIN
	UNDERGROUND TELEPHONE/COMMUNICATIONS
	FENCE
	INVISIBLE LINE ABOVE
	INVISIBLE LINE BELOW
	LINE to be REMOVED
	CENTER LINE
	FINISH FLOOR, BEARING, or BUILDING LINE
	GRID LINE
	DOOR NUMBER
	WINDOW NUMBER KEY (on Exterior Elevations)
	INTERIOR ELEVATION NUMBER
	TEMPERED GLASS REQUIRED
	MATCH LINE
	DETAIL KEY
	DETAIL NUMBER/TITLE
	BUILDING SECTION KEY (With Sheet Number)
	WALL SECTION KEY (With Sheet Number)
	ROOM NUMBER (Elev. Number With Sheet Number) (Name on Tail)
	NOTE KEY
	PARTITION KEY
	ASPHALTIC CONCRETE PAVEMENT (As Noted)
	CONCRETE WALK OR SURFACE (As Noted)
	MASONRY SURFACE (As Noted)
	CONCRETE WALL (PLAN)
	BRICK MASONRY WALL (PLAN)
	CONCRETE MASONRY UNIT WALL
	WOOD FRAME WALL (PLAN)
	MAIN BEARING POINT in WALL (Member as Indicated)
	RATED WALL
	METAL FRAME WALL (PLAN)
	EIFS/MTL FRAME WALL (PLAN)
	MTL BUILDING WALL (PLAN)
	WOOD FRAMING MEMBER (Nominal Size Noted)
	WOOD BLOCKING MEMBER (Nominal Size Noted)
	WOOD FINISH MEMBER (Net Size Noted)
	PLYWOOD / PARTICLE BOARD
	SOLID ACRYLIC POLYMER
	INSULATION/EIFS, RIGID
	INSULATION, FOAMED IN PLACE
	METAL
	GYPSUM BD / PLASTER / CEMENT BOARD
	INSULATION, BATT AND/OR LOOSE FILL
	TILE
	ASPHALTIC PAVEMENT
	CONCRETE
	EARTH (SECTION)
	ACCESS CONTROL READER
	EXIT SIGN
	FLOURESCENT LIGHTING
	PENDANT/CAN LIGHT
	MECHANICAL DIFFUSER
	ELECTRICAL POWER POLE
	FIRE SUPPRESSION HEAD (IN ACT)
	SPEAKER
	LIGHTING CONTROLS (IN ACT)

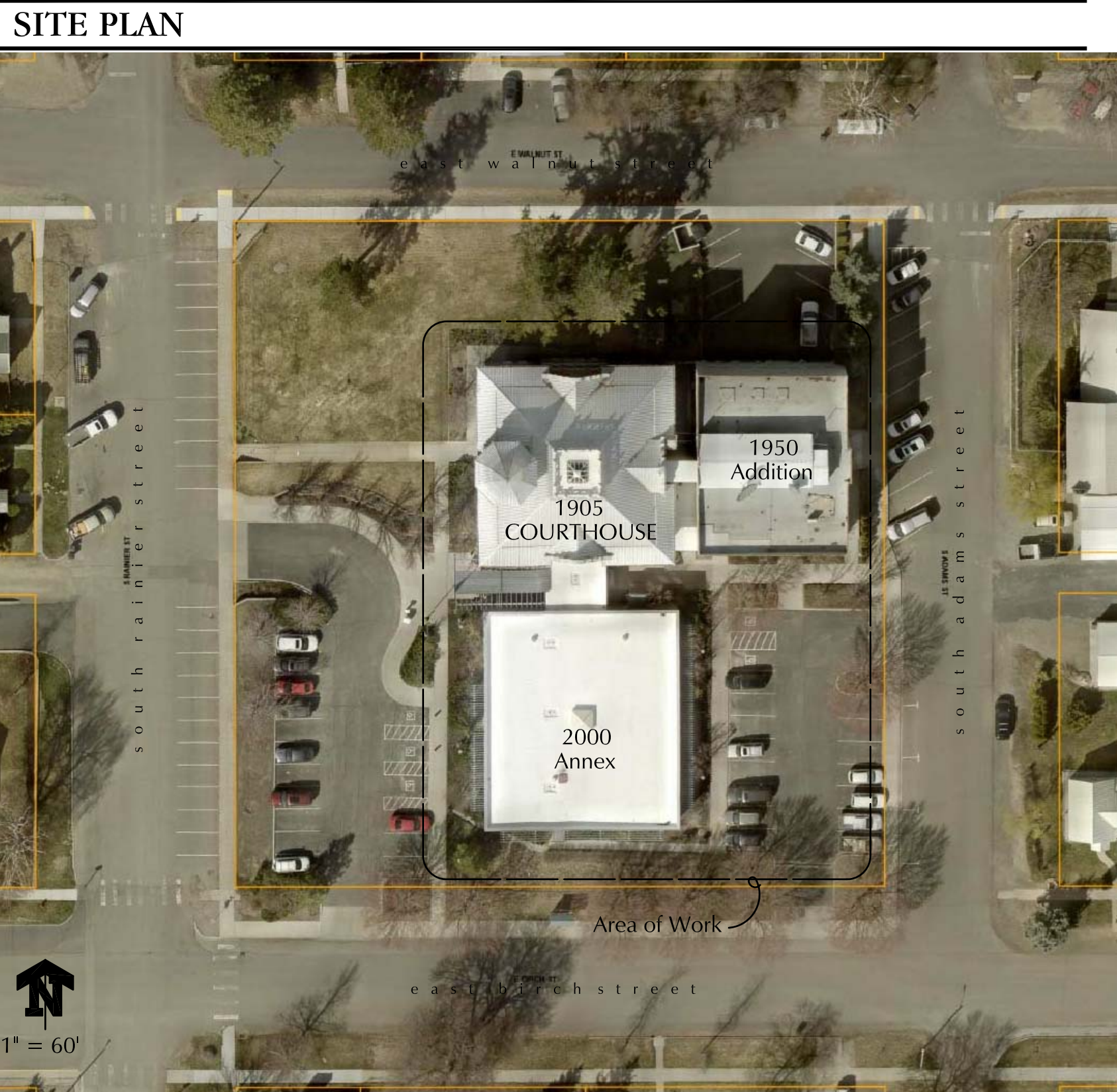
- ### GENERAL NOTES
- ALL WORK SHALL CONFORM TO ALL APPLICABLE CODES AND ORDINANCES, THE MORE STRINGENT IS TO GOVERN; DISCREPANCIES BETWEEN CONTRACT DOCUMENTS AND CODES SHALL BE BROUGHT TO THE ARCHITECT'S ATTENTION PROMPTLY AND A RESOLUTION OBTAINED BEFORE PROCEEDING.
 - DRAWINGS ARE NOT TO BE SCALED. DIMENSIONAL DATA SHALL BE OBTAINED FROM WRITTEN INFORMATION ONLY, OR AS-BUILT MEASUREMENTS. VERIFY ALL DIMENSIONS BEFORE PROCEEDING. ANY DIMENSIONAL DEVIATION FROM THAT SHOWN ON THE DRAWINGS, WHICH MAY AFFECT THE INTENT OF THE DESIGN OR PROPER INCORPORATION OF ELEMENTS, SHALL BE BROUGHT TO THE ARCHITECT'S ATTENTION PROMPTLY AND A RESOLUTION OBTAINED BEFORE PROCEEDING.
 - THE ARCHITECT SHALL BE INFORMED IMMEDIATELY OF ANY DISCREPANCY BETWEEN THE CONTRACT DOCUMENTS AND THE SITE CONDITIONS.
 - THE TERM "FINISHED FLOOR (FIN. FLR. or F.F.)" REFERS TO THE TOP OF FINISHED SLAB WHERE CONCRETE FLOOR OCCUR AND TOP OF FLOOR SHEATHING AT WOOD STRUCTURES.
 - EXTERIOR DIMENSIONS ARE TO THE FACE OF CONCRETE / SHEATHING UNLESS OTHERWISE NOTED. INTERIOR DIMENSIONS ARE TO THE FACE OF FRAMING UNLESS INDICATED AS A CENTERLINE OR SPECIFICALLY NOTED OTHERWISE. CLEAR DIMENSIONS SHALL BE FROM FINISH TO FINISH.
 - NOTATIONS OR DETAILS KEYED TO VARIOUS DRAWING SYMBOLS, PATTERNS, ETC. SHALL APPLY TYPICALLY TO ALL SIMILARLY INDICATED ITEMS, LOCATIONS, OR CONDITIONS NOT OTHERWISE KEYED.
 - PRESERVE AND PROTECT EXISTING UTILITIES AND BUILDING COMPONENTS WHICH MAY BE PRESENT AND ARE NOT SCHEDULED OR REQUIRED TO BE CHANGED OR REMOVED.
 - ADJUST NEW CONSTRUCTION TO ALIGN WITH EXISTING CONSTRUCTION SUCH THAT FINISHES MAY BE APPLIED ALONG STRAIGHT AND TRUE LINES, UNLESS SPECIFICALLY NOTED OTHERWISE.

DESIGN CRITERIA		
SITE DATA:		
Governing Jurisdiction:	Town of Waterville	
Zone:	PUZ (Public Use)	
Site Area:	1.58 Acres	
Lot Coverage:	12.1% (40% Max)	
Setbacks:	25' Front, minimum	
	10' Side, minimum	
	15' Rear, minimum	
Building Height:	50' maximum	
BUILDING DATA :		
Occupancy Groups:	B/A-3 nonseparated	
Building Construction Type:	Ⅱ-B sprinkled	
Occupant Load:	416	
Allowable Building Height:	60' (3 stories)	
Actual Building Height:	±105' (3 Story)	
Allowable Building Area:	22,500 S.F. (18,000 +6,000 x 0.75 for 30' yards all sides)	
Actual Enclosed Floor Area:	35,992 S.F.	
	17,012 S.F.	
	9,490 S.F.	
	9,490 S.F.	

PROJECT INFORMATION	
JOB ADDRESS:	LEGAL DESCRIPTION:
DOUGLAS COUNTY COURTHOUSE	LOTS 1 THRU 20; VAC 16' WALNUT ST, VAC 10' ADAMS ST & VAC ALLEY; KINCAID'S 1ST ADD
203 S RAINIER STREET	
WATERVILLE, WA 98858	
BUILDING OWNER:	PARCEL NUMBER
DOUGLAS COUNTY	11901800100
140 19th STREET NW	
EAST WENATCHEE, WA 98802	



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E0.05	Electrical Panel Schedules		



REGISTERED ARCHITECT
PAUL R. COFFOCK
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DRW - CHKS -

Douglas County Commissioners

COUNTY COURTHOUSE

Heating, Ventilation & Air Conditioning Improvements

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Job: 2241 Date: 2/15/24

DWG ID - 2242 0101-.dwg

A1.1

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LIST OF DRAWINGS	
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M4.08	MECHANICAL DETAILS
M4.09	REFRIGERANT DIAGRAMS

MECHANICAL LEGEND			
SYMBOL	DESCRIPTION	ABBREV.	DESCRIPTION
— C —	CONDENSATE LINE	AAV	AUTOMATIC AIR VENT
— RG —	REFRIGERANT GAS (RG)	AFF	ABOVE FINISHED FLOOR
— RL —	REFRIGERANT LIQUID (RL)	APPROX	APPROXIMATELY
— CHS —	CHILLED WATER SUPPLY (CHS)	ARCH	ARCHITECTURAL
— CHR —	CHILLED WATER RETURN (CHR)	ASSY	ASSEMBLY
	ISOLATION VALVE	BLDG	BUILDING
	COMBINATION SHUT-OFF & BALANCING VALVE	B.O.D.	BOTTOM OF DUCT
	CHECK VALVE	BTU	BRITISH THERMAL UNIT
	UNION	BTUH	BRITISH THERMAL UNIT/HOUR
	PIPING FLEXIBLE CONNECTOR	CAP	CAPACITY
	RELIEF VALVE OR SAFETY VALVE	CFM	CUBIC FEET PER MINUTE
	MANUAL AIR VENT, AAV = AUTOMATIC AIR VENT	CLG	CEILING
	STRAINER	CTG	CEILING TRANSFER GRILLE
	CONCENTRIC REDUCER	CO	CLEANOUT
	PRESSURE GAUGE	COMP	COMPRESSOR
	THERMOMETER	C	CONDENSATE
	DUCT (FIRST FIGURE, SIDE SHOWN)	CONN	CONNECTION
	LINED DUCT (DIM. FOR NET FREE AREA)	CONT	CONTINUE, CONTINUATION
	RISE (R) OR DROP (D) ARROW IN DIRECTION OF FLOW	COP	COEFFICIENT OF PERFORMANCE
	VOLUME DAMPER (MANUAL)	CW	COLD WATER
	MOTORIZED DAMPER	DB	DRY BULB
	FLEXIBLE CONNECTION	DEG F, F	DEGREE FAHRENHEIT
	FLEXIBLE DUCT	DIA, Ø	DIAMETER
	CEILING OUTLET	DN	DOWN
	CEILING INLET	DWG	DRAWING
	WALL OUTLET (OR INLET)	EA	EACH
	THERMOSTAT	ECM	ELECTRONICALLY COMMUTATED MOTOR
		ELEC	ELECTRICAL, ELECTRIC
		EER	ENERGY EFFICIENCY RATIO
		EAT	ENTERING AIR TEMPERATURE
		EWB	ENTERING WET BULB
		EDB	ENTERING DRY BULB
		EOL	END OF LINING
		ER	ENERGY RECOVERY VENTILATOR
		EXH	EXHAUST
		EXIST. (E)	EXISTING
		ESP	EXTERNAL STATIC PRESSURE
		FV	FACE VELOCITY
		FPM	FEET PER MINUTE
		FA	FIRE ALARM
		FDC	FIRE DEPARTMENT CONNECTION
		FLEX	FLEXIBLE
		FL	FLOOR
		FCO	FLOOR CLEAN OUT
		FLA	FULL LOAD AMPS
		FSD	FIRE SMOKE DAMPER
		GAL	GALLON
		GALV.	GALVANIZED
		G.C.	GENERAL CONTRACTOR
		GPM	GALLON PER MINUTE
		GPH	GALLON PER HOUR
		HSPF	HEATING SEASON PERFORMANCE FACTOR
		HP	HORSE POWER
		HRU	HEAT RECOVERY UNIT
		HW	HOT WATER
		HWC	HOT WATER CIRCULATING
		HWR	HOT WATER RETURN
		HWS	HOT WATER SUPPLY
		IN	INCH
		I.E.	INVERT ELEVATION
		KW	KILOWATT
		LAT	LEAVING AIR TEMPERATURE
		LDB	LEAVING DRY BULB
		LWT	LEAVING WATER TEMPERATURE
		LWB	LEAVING WET BULB
		MAX	MAXIMUM
		MBH	THOUSAND BTUH
		MCA	MINIMUM CIRCUIT AMPS
		MECH	MECHANICAL
		MD	MOTORIZED DAMPER
		MFR	MANUFACTURER
		MFS	MAXIMUM FUSE SIZE
		MIN	MINIMUM
		NC	NORMALLY CLOSED
		NO	NORMALLY OPEN
		NO.	NUMBER
		NTS	NOT TO SCALE
		OA	OUTSIDE AIR
		OBD	OPPOSED BLADE DAMPER
		PD	PRESSURE DROP
		PH	PHASE
		RA	RETURN AIR
		REF	REFERENCE
		RG	REFRIGERANT GAS
		RL	REFRIGERANT LIQUID
		REL	RELIEF
		REQ'D	REQUIRED
		RLA	RATED LOAD AMPS
		RPM	REVOLUTIONS PER MINUTE
		RM	ROOM
		SA	SUPPLY AIR
		S.P.	SCREENED OPENING
		S.S.	STAINLESS STEEL
		STD	STANDARD
		TEMP	TEMPERATURE
		TD	TRANSFER DUCT
		TG	TRANSFER GRILLE
		TSP	TOTAL STATIC PRESSURE
		T'STAT	THERMOSTAT
		TYP	TYPICAL
		UNO	UNLESS NOTED OTHERWISE
		V	VOLTS, VOLTAGE - VENT
		VRF	VARIABLE REFRIGERANT FLOW
		VFD	VARIABLE FREQUENCY DRIVE
		VTR	VENT THROUGH ROOF
		W	WATT
		W/	WITH
		WB	WET BULB
		WCO	WALL CLEAN OUT - WASTE
		WSEC	WASHINGTON STATE ENERGY CODE
	DETAIL IDENTIFICATION NUMBER SHEET ON WHICH DETAIL IS SHOWN		
	SECTION IDENTIFICATION LETTER SHEET ON WHICH SECTION IS SHOWN		

M0.01



ENERGY CODE NOTES					
EQUIPMENT SIZING, PERFORMANCE, AND TYPE			ECONOMIZER		
1. LOAD CALCULATIONS, C403.1.2: LOAD CALCULATIONS HAVE BEEN PERFORMED IN ACCORDANCE WITH WSEC.	17. AUTOMATIC START: AUTOMATIC START CONTROLS SHALL BE PROVIDED FOR EACH HVAC SYSTEM, AND BE CAPABLE OF AUTOMATICALLY ADJUSTING DAILY START TIME IN ORDER TO BRING EACH SPACE TO THE DESIRED OCCUPIED TEMPERATURE IMMEDIATELY PRIOR TO SCHEDULED OCCUPANCY.		25. AIR ECONOMIZER REQUIRED, C403.5: SEE EQUIPMENT SCHEDULES FOR EQUIPMENT PROVIDED WITH ECONOMIZER COOLING.		
2. EQUIPMENT AND SYSTEM SIZING, C403.3: OUTPUT CAPACITIES OF HEATING AND COOLING EQUIPMENT AND SYSTEMS ARE NO GREATER THAN THE SMALLEST AVAILABLE EQUIPMENT SIZE THAT EXCEEDS THE CALCULATED LOADS.	18. OUTDOOR AIR DAMPERS: OUTSIDE AIR INTAKE DAMPERS SHALL AUTOMATICALLY CLOSE WHEN SYSTEM OR SPACES SERVED ARE NOT IN USE OR DURING WARM-UP AND SET BACK.		26. INTEGRATED ECONOMIZER OPERATION, C403.5: ECONOMIZERS SHALL BE CONFIGURED FOR PARTIAL COOLING OPERATION EVEN WHERE ADDITIONAL MECHANICAL COOLING IS REQUIRED TO MEET THE LOAD.		
3. HVAC EQUIPMENT PERFORMANCE, C403.3.2: EQUIPMENT SCHEDULES ARE INCLUDED WITH THESE PLANS. GAS FIRED EQUIPMENT SHALL HAVE INTERMITTENT IGNITION.	19. VENTILATION: MECHANICAL VENTILATION AIR SYSTEMS SHALL BE CONFIGURED TO PROVIDE NOT MORE THAN 150%, BUT AT LEAST THE MINIMUM REQUIRED VOLUME OF OUTDOOR AIR TO EACH ZONE PER IMC. SEE MECHANICAL EQUIPMENT SCHEDULES FOR MINIMUM OUTSIDE AIR VALUES.		27. ECONOMIZER HEATING IMPACT: ECONOMIZER CONTROLS SHALL BE SUCH THAT ECONOMIZER OPERATION DOES NOT INCREASE BUILDING HEATING ENERGY DURING NORMAL OPERATIONS.		
4. ELECTRIC MOTOR EFFICIENCY, C405.8: ALL ELECTRIC MOTORS SHALL MEET THE MINIMUM EFFICIENCY OF WSEC. FRACTION HP FAN MOTORS 1/12HP OR GREATER UP TO 1HP SHALL BE ECM TYPE OR SHALL HAVE A MINIMUM EFFICIENCY OF 70% OR GREATER.	DUCTING SYSTEMS		28. AIR ECONOMIZER CAPACITY: ECONOMIZER OUTSIDE AIR AND RETURN AIR DAMPERS SHALL MODULATE TO PROVIDE UP TO 100% OUTSIDE AIR FOR COOLING.		
5. FAN POWER LIMITATION, C403.8.1: FOR ALL HVAC SYSTEMS WITH TOTAL FAN HP > 5HP, MOTOR HP OR BHP SHALL COMPLY WITH FAN POWER LIMITATIONS PER EQUATIONS IN TABLE C403.2.11.1(1).	20. DUCT CONSTRUCTION, C403.10.2: DUCTWORK SHALL BE CONSTRUCTED AND SEALED PER IMC.		29. INTEGRATED AIR ECONOMIZER: MECHANICAL COOLING CONTROLS SHALL BE INTERLOCKED WITH AIR ECONOMIZER CONTROLS SO THE OUTSIDE AIR DAMPER REMAINS IN 100% OPEN POSITION WHEN MECHANICAL EQUIPMENT IS IN COOLING TO MEET THE COOLING LOAD, UNTIL THE LEAVING AIR TEMPERATURE IS < 45°F. FOR SYSTEMS WITH COOLING CAPACITY GREATER THAN 65 MBH, CONTROL OF ECONOMIZER DAMPERS SHALL NOT BE CONTROLLED BASED ONLY ON MIXED AIR TEMPERATURE.		
6. MOTOR NAMEPLATE HP, C403.8.2: FOR EACH FAN, THE MOTOR SHALL BE NO LARGER THAN THE FIRST AVAILABLE MOTOR SIZE GREATER THAN THE BHP.	21. DUCT PRESSURE CLASS, C403.10.2: ALL DUCTWORK SHOWN IS LOW PRESSURE DUCT, OPERATING AT STATIC PRESSURE LESS THAN OR EQUAL TO 2 INCH WATER GAUGE (W.G.).		30. ECONOMIZER HIGH LIMIT CONTROLS: PROVIDE HIGH-LIMIT ECONOMIZER SHUT-OFF CONTROL UTILIZING FIXED DRY BULB METHOD. ECONOMIZER SHALL BE OFF WHEN OUTDOOR AIR TEMPERATURE EXCEEDS 75°F. PROVIDE WITH FAULT PROTECTION PER WSEC.		
7. FAN EFFICIENCY, C403.8.1: FANS AND FAN SYSTEMS GREATER THAN 5HP SHALL HAVE A FAN EFFICIENCY GRADE (FEG) OF 67 OR HIGHER PER AMCA 205.	22. DUCT INSULATION, C403.10.1: MINIMUM DUCT INSULATION PER WSEC IS AS FOLLOWS:		COMMISSIONING / BALANCING		
8. LARGE CAPACITY BOILER SYSTEMS, C403.4.3: HYDRONIC SYSTEMS WITH A SINGLE BOILER OVER 500 MBH INPUT CAPACITY SHALL HAVE MULTIPLE STAGES OR MODULATION CAPACITY. BOILER SYSTEMS OVER 1,000 MBH SHALL HAVE THE FOLLOWING MINIMUM TURNDOWN RATIO OF 3:1.	SERVICE	INSULATION LEVEL	31. PROJECT'S HVAC SYSTEMS ARE TO BE COMMISSIONED PER PROJECT SPECIFICATIONS.		
9. PACKAGED ELECTRIC EQUIPMENT, C403.2.4: ALL PACKAGED ELECTRIC HEATING AND COOLING EQUIPMENT WITH TOTAL COOLING CAPACITY GREATER THAN 6,000 BTU/H SHALL BE A HEAT PUMP CONTROLLED BY OCCUPANCY DEVICE OR TIME SCHEDULE.	OUTSIDE AIR DUCTS AND PLENUMS TO	PROVIDE INSULATION EQUIVALENT	32. ALL SYSTEMS ARE TO BE BALANCED PER C408.2.2 & PROJECT SPECIFICATIONS.		
10. OUTDOOR AIR, EXHAUST & RELIEF DAMPERS: PROVIDE ALL OUTSIDE AIR, EXHAUST AIR, AND RELIEF AIR OPENINGS WITH CLASS 1 (MAX LEAKAGE OF 4 CFM/SF AT 1.0" W.C.) MOTORIZED DAMPERS.	METAL	ENVELOPE REQUIREMENT FOR FRAMED WALLS	PROJECT CLOSE OUT DOCUMENTATION		
11. RETURN AIR DAMPERS: PROVIDE RETURN AIR OPENINGS WITH CLASS 1 MOTORIZED DAMPER WHERE USED FOR AIRSIDE ECONOMIZER.	OUTSIDE AIR DUCT SERVING INDIVIDUAL SUPPLY UNIT	R-8	33. DOCUMENTATION SUBMITTAL REQUIREMENTS, C103.6: SUBMIT ALL CLOSEOUT DOCUMENTATION INCLUDING AS-BUILTS AND O&M'S TO OWNER.		
	SUPPLY & RETURN DUCTS IN UNCONDITIONED SPACES	R-6	34. THESE "ENERGY CODE NOTES" ARE LISTED TO SATISFY THE BUILDING DEPARTMENT'S REQUIREMENT THAT CERTAIN INFORMATION BE PLACED ON THE PLANS, BUT DO NOT DIMINISH THE FULL PROJECT REQUIREMENTS. PROVIDE ITEMS IN EXCESS OF CODE WHERE NOTED ON DRAWINGS AND IN SPECIFICATIONS. FOR OTHER ADDED REQUIREMENTS, SEE SPECIFICATIONS.		
	SUPPLY DUCTS WITHIN CONDITIONED SPACE WHERE SUPPLY AIR IS < 55 DEG F. OR > 105 DEG F.	R-3.3			
HVAC SYSTEM CONTROLS					
12. THERMOSTATIC CONTROLS: WHERE ADJACENT ZONES CONNECTED BY PERMANENT OPENINGS WITH AREA GREATER THAN 10% OF EITHER ZONE SF, PROVIDE CONTROL TO PREVENT ADJACENT ZONES FROM OPERATING IN CONFLICTING MODES. WHERE A NON-PERIMETER ZONE IS ADJACENT TO A PERIMETER ZONE, PROVIDE CONTROLS TO ONLY ALLOW COOLING IN THE NON-PERIMETER ZONE WHEN IT IS 5 DEGREES HIGHER THAN THE PERIMETER ZONE.	PIPING SYSTEMS				
13. HEAT PUMP SUPPLEMENTAL HEAT, C403.4.1.1: HEAT PUMPS SHALL USE COMPRESSION AS FIRST STAGE OF HEATING AND SUPPLEMENTAL HEATING AS SECOND STAGE CONTROLLED SO AS TO PREVENT OPERATION ABOVE 40°F.	23. PIPING INSULATION, C403.10.3: MINIMUM PIPE INSULATION PER WSEC IS AS FOLLOWS:				
14. DEADBAND: THERMOSTATIC CONTROLS SHALL BE CONFIGURED WITH 5°F MINIMUM DEADBAND FOR SYSTEMS THAT CONTROL BOTH HEATING AND COOLING.	FLUID OPERATING	INSULATION THICKNESS (NOMINAL PIPE SIZE)			
15. SETPOINT OVERLAP RESTRICTION: WHERE SEPARATE HEATING AND COOLING SYSTEMS WITH SEPARATE THERMOSTATIC CONTROL DEVICES SERVE A ZONE, PROVIDE A LIMIT SWITCH, MECHANICAL STOP, OR DDC CONTROL TO PREVENT SIMULTANEOUS HEATING AND COOLING.	TEMPERATURE	<1 1 TO <1-1/2 1-1/2 TO <4 4 TO <8 OVER 8			
16. AUTOMATIC SETBACK AND SHUTDOWN: HVAC SYSTEMS SHALL BE EQUIPPED WITH AUTOMATIC CONTROLS CAPABLE OF STARTING AND STOPPING THE SYSTEM FOR SEVEN DIFFERENT DAILY SCHEDULES, AND SHALL HAVE MANUAL OVERRIDE CONFIGURED TO OPERATE THE SYSTEM FOR 2 HOURS.	>350	4.5 5.0 5.0 5.0 5.0			
	251-350	3.0 4.0 4.5 4.5 4.5			
	201-250	2.5 2.5 2.5 3.0 3.0			
	141-200	1.5 1.5 2.0 2.0 2.0			
	105-140	1.0 1.0 1.5 1.5 1.5			
	40-60	0.5 0.5 1.0 1.0 1.0			
	<40	0.5 1.0 1.0 1.0 1.5			
	24. PIPE INSULATION EXPOSED TO WEATHER, C403.10.3.1: PROVIDE METAL JACKETING ON ALL PIPE INSULATION EXPOSED TO WEATHER AND SEAL ALL SEAMS WATER TIGHT.				

HEAT PUMP SCHEDULE - SPLIT SYSTEM TYPE																															
SYMBOL	BASIS OF DESIGN MANUFACTURER AND SERIES NO.	AREA SERVED	INDOOR UNIT TYPE	COOLING CAP. *			HEATING CAP. **		A - INDOOR UNIT ***							B - OUTDOOR UNIT ***							FILTERS		MIN. OA CFM	MAX. OUTDOOR UNIT WEIGHT	MAX. INDOOR UNIT WEIGHT	REFRIGERANT PIPE SIZE		REMARKS	
				TOTAL MBH	SENS. MBH	EFFICIENCY	MBH	EFFICIENCY	FAN			HEATER		TOTAL ELECTRIC		COMPRESSOR		FAN		ELECTRICAL			TYPE	MIN. SF				RG	RL		
									CFM	ESP	FLA	KW	STAGES	MCA	MOP	VOLTS/PH	QTY	RLA (EA)	QTY	FLA	MCA	MOP									VOLTS/PH
HP-10	TRANE TPVAA030 TRUZH030	FIRST FLOOR SHERIFF'S	DUCTED AIR HANDLER	30	22	18 SEER 2	32	3.8 COP	730	0.5"	3.3	3	1	40.2	45	208/1	1	9	1	0.5	24	40	208/1		-	300	150	5/8"	3/8"	③	
HP-20	MITSUBISHI SUZ-KA12 SLZ-KF12	FIRST FLOOR MAINT OFFICE	CEILING CASSETTE	15	8	20.3 SEER	17	3.4 COP	230	-	-	-	-	0.3	-	-	1	10	1	0.7	14	24	208/1	MFR STD	MFR STD	-	150	40	3/8"	1/4"	①②
* COOLING CAPACITY IS AHRI RATING: AT 80° F DB; 67° F WB INDOOR COIL EAT AND 95° F OUTDOOR COIL EAT AT UNIT NOMINAL CFM. ** HEATING CAPACITY IS AHRI HI-TEMP RATING: AT 70° F DB INDOOR EAT AND 47° F DB; 43° F WB OUTDOOR COIL EAT. *** ON PLANS "A" DESIGNATES INDOOR UNIT, "B" DESIGNATES OUTDOOR UNIT. (E.G. HP-1B IS HP-1 OUTDOOR UNIT). NOTES: 1. HEATING CAPACITY AT OA TEMPERATURE OF -13°F SHALL BE NO LESS THAN 80% OF SCHEDULED HEATING CAPACITY. 2. PROVIDE INDOOR UNITS W/ CONDENSATE PUMPS AS REQUIRED FOR CONDENSATE DRAINAGE. 3. PROVIDE OUTDOOR UNITS W/ BASE PAN HEATER, SNOW/WIND SHIELD. 4. HEATER RATINGS ARE AT SCHEDULED VOLTAGE. ① INDOOR UNIT POWERED FROM OUTDOOR UNIT. ② INDOOR UNIT TO BE SIZED FOR INSTALLATION IN 2'x2' T-BAR. ③ PROVIDE INDOOR UNIT W/ SINGLE POINT POWER CONN & ELECTRICAL DISCONNECT. ④ FILTER(S) IN CEILING RA GRILLE(S).																															

CHILLER SCHEDULE																			
SYMBOL	BASIS OF DESIGN MANUFACTURER AND SERIES NUMBER	TYPE	COOLING PERFORMANCE							COMPRESSORS		FANS			ELECTRICAL			MAX WEIGHT LBS	REMARKS
			CAP TONS	EFF	EWT	LWT	GPM	WATER PD, FT	OAT	QTY	RLA EA.	QTY	RLA EA.	KW EA.	MCA	MOP	VOLTS/PH		
CH-1	TRANE CGAM070F2	PACKAGED AIR COOLED SCROLL	68.5	11 EER 16.27 IPLV	54	44	164	20	95	4	(2) 33 (2) 25	6	3.2	1.23	145	200	460/3	11,000	R-454B REFRIGERANT, W/ ACOUSTIC WRAP, 2 REFRIG CIRCUIT

NOTES:

1. UNIT A-WEIGHTED SOUND POWER SHALL NOT EXCEED 91 dBA AT FULL LOAD, DESIGN CONDITIONS, PER AHRI.

2. UNIT SHALL BE FOR USE WITH 35% PROPYLENE GLYCOL; SCHEDULED PERFORMANCE IS WITH 0% GLYCOL.

3. PROVIDE W/ TRANSFORMER TO SERVE UNIT CONTROL CIRCUIT.

4. COMPRESSOR & FAN QTY SHOWN IS MINIMUM.

MISCELLANEOUS SCHEDULE								
SYMBOL	BASIS OF DESIGN MANUFACTURER AND SERIES NO.	TYPE	AREA / UNIT SERVED	NOMINAL SIZE	CAPACITY	ELECTRICAL		REMARKS
						WATTS	VOLTS/PH	
EH-20	KING LPWC	CEILING ELECTRIC HEATER	MAINT OFFICE	-	-	2250	208/1	W/ WALL MOUNT PROGRAMMABLE THERMOSTAT, T-BAR CEILING ADAPTER
EH-30	INDEECO QUA	DUCT ELECTRIC HEATER	DOAS-3	8x8	-	2500	208/1	W/ SA THERMOSTAT, OA LOCKOUT, & SCR CONTROL
DOAS-30	RENEWAIRE EV PREMIUM M	ERV	COMMISSIONER'S AREA	-	200 CFM @ 0.5" ESP	190	115/1	ENERGY RECOVERY PER WSEC, W/ MERV 8 FILTERS

FAN SCHEDULE													
SYMBOL	BASIS OF DESIGN MANUFACTURER AND SERIES NO.	TYPE	AREA SERVED	CFM	ESP	MAX RPM	BRAKE HP	ELECTRICAL		DRIVE	CONTROL	MAX WEIGHT LBS	REMARKS
								HP	VOLTS / PH				
REF-1	COOK 270-SQN	INLINE	RELIEF	7300	0.75"	1300	2.8	3	208/3	DIRECT	DUCT SP	260	MAX INLET 17.4 SONES W/ VFD
REF-2	COOK 180-TLCD	ROOFTOP LOW PROFILE	RELIEF	2900	0.62"	1200	0.63	0.75	208/1	DIRECT	DUCT SP	150	MAX INLET 21.0 SONES W/ ECM
NOTES: 1. PROVIDE FANS WITH EC MOTORS (MOTOR MOUNTED) OR VARIABLE FREQUENCY DRIVES. 2. BALANCE RF'S FOR 95% OF SCHEDULED CFM TO ALLOW FOR BLDG PRESSURIZATION. 3. PROVIDE REF FANS WITH MOTORIZED DAMPERS TO BE INTERLOCKED WITH FANS; LOCATE AS SHOWN ON PLANS. 4. PROVIDE ALL ROOFTOP FANS WITH CURBS.													

AIR INLET & OUTLET SCHEDULE			
SYMBOL	TYPE	MANUFACTURER AND SERIES NUMBER	REMARKS
CD	CEILING DIFFUSER	KRUEGER SERIES 1240	MODULAR CORE, SQUARE NECK
CRG	CEILING RETURN GRILLE	KRUEGER SERIES EGC-5	1/2"x1/2"x1/2" CUBE CORE
FRG	FILTER RETURN GRILLE	KRUEGER SERIES EGC-5FF	1/2"x1/2"x1/2" CUBE CORE W/ 2" FILTER FRAME, MERV 8
 VG	VENTILATION GRILLE	LIFEBREATH TECHGRILLE	

NOTES:

- CEILING DIFFUSERS (CD) SHALL HAVE NO. & DIRECTION OF THROWS AS INDICATED ON PLANS.
(E.G. CD-3 = 3 WAY THROW)
- ALL AIR TERMINALS SHALL HAVE FACTORY FINISH, COLOR WHITE.
- SEE LEGEND FOR TERMINOLOGY USED IN AIR TERMINAL CALL-OUTS ON DRAWINGS.
- FIELD VERIFY (E) CEILING TYPES, PROVIDE AIR TERMINALS TO MATCH CEILING CONSTRUCTION INSTALLED IN.
- TRANSFER GRILLES (CTG'S) SHALL BE SAME AS CRG'S.



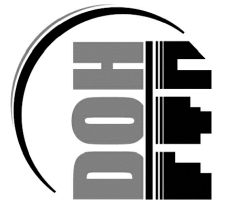
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ARCHITECTS and PLANNERS



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VRF OUTDOOR HEAT PUMP SCHEDULE														
SYMBOL	BASIS OF DESIGN MANUFACTURER AND SERIES NO.	OUTDOOR UNIT COMBINATION	AREA SERVED	COOLING CAP. *		HEATING CAP.**		OUTDOOR FAN	COMPRESSOR	MAX. WEIGHT LBS	UNIT ELECTRICAL			REMARKS
				TOTAL MBH	EFF (EER / IEER)	MBH	EFF (COP)				QTY	QTY	MCA	
HP-30	TRANE R2-SERIES TURYH0964AN40AN	-	JURY COMM OFFICES	96	13.7 / 26.5	108	3.94	1	1	700	20	30	460/3	"HYPER HEAT" MODEL
* COOLING CAPACITY IS AHRI RATING: AT 80°F DB; 67°F WB INDOOR COIL EAT AND 95°F OUTDOOR COIL EAT. ** HEATING CAPACITY IS AHRI HI-TEMP RATING: AT 70°F DB INDOOR COIL EAT AND 47°F DB; 43°F WB OUTDOOR COIL EAT. UNIT SHALL BE ABLE TO PROVIDE FULL HEAT OUTPUT DOWN TO -4 DEG F.														

VRF INDOOR HEAT PUMP SCHEDULE													
SYMBOL	BASIS OF DESIGN MANUFACTURER AND SERIES NO.	TYPE	AREA SERVED	SERVED BY	CAPACITY (MBH)		CFM	ESP	FILTERS	UNIT ELECTRICAL		MAX WEIGHT	REMARKS
					COOLING	HEATING				MCA	VOLT/PH		
BC-1	TRANE TCMB	BC CONTROLLER	HP-30	-	SEE HP-30 CAPACITY		-	-	-	1	208/1	130	
CC-8-1	TRANE TPLFY-P08NFM4	CEILING CASSETTE	COMM OFFICE INTERIOR	HP-30	8	9	315	-	①	2.8	208/1	40	③
CC-15-2	TRANE TPLFY-EP15NEM4	CEILING CASSETTE	COMM OFFICE ENTRY	HP-30	15	17	500	-	①	2.8	208/1	60	
CC-15-3	TRANE TPLFY-EP15NEM4	CEILING CASSETTE	COMM OFFICE MEETING	HP-30	15	17	500	-	①	2.8	208/1	60	
CC-8-4	TRANE TPLFY-P08NFM4	CEILING CASSETTE	COMM OFFICE NE OFFICE	HP-30	8	9	280	-	①	2.8	208/1	40	③
CC-15-5	TRANE TPLFY-EP15NEM4	CEILING CASSETTE	COMM OFFICE SE OFFICE	HP-30	15	17	500	-	①	2.8	208/1	60	
CC-15-6	TRANE TPLFY-EP15NEM4	CEILING CASSETTE	COMM OFFICE SOUTH OFFICE	HP-30	15	17	500	-	①	2.8	208/1	60	
CC-15-7	TRANE TPLFY-EP15NEM4	CEILING CASSETTE	COMM OFFICE SW OFFICE	HP-30	15	17	500	-	①	2.8	208/1	60	
AH-30-8	TRANE TPEFYP030MA	AIR HANDLER	JURY ROOM	HP-30	30	34	860	0.3	②	4.2	208/1	150	
CC-8-9	TRANE TPLFY-P08NFM4	CEILING CASSETTE	JURY HALLWAY	HP-30	8	9	280	-	①	2.8	208/1	40	
NOTES: 1. SUBMIT DRAWINGS OF FINAL REFRIGERATION PIPING ARRANGEMENT; SEE SPECIFICATIONS FOR FULL REQUIREMENTS.									①	MFR'S STANDARD FILTER, INTEGRAL TO UNIT.			
2. PROVIDE UNITS W/ CONDENSATE PUMPS AS REQ'D TO ALLOW FOR PROPER CONDENSATE DRAIN.									②	FILTER(S) IN CEILING RA GRILLE(S).			
									③	FOR INSTALLATION IN 24x24 T-BAR CEILING MODULE.			



MECHANICAL CONNECTION NOTES

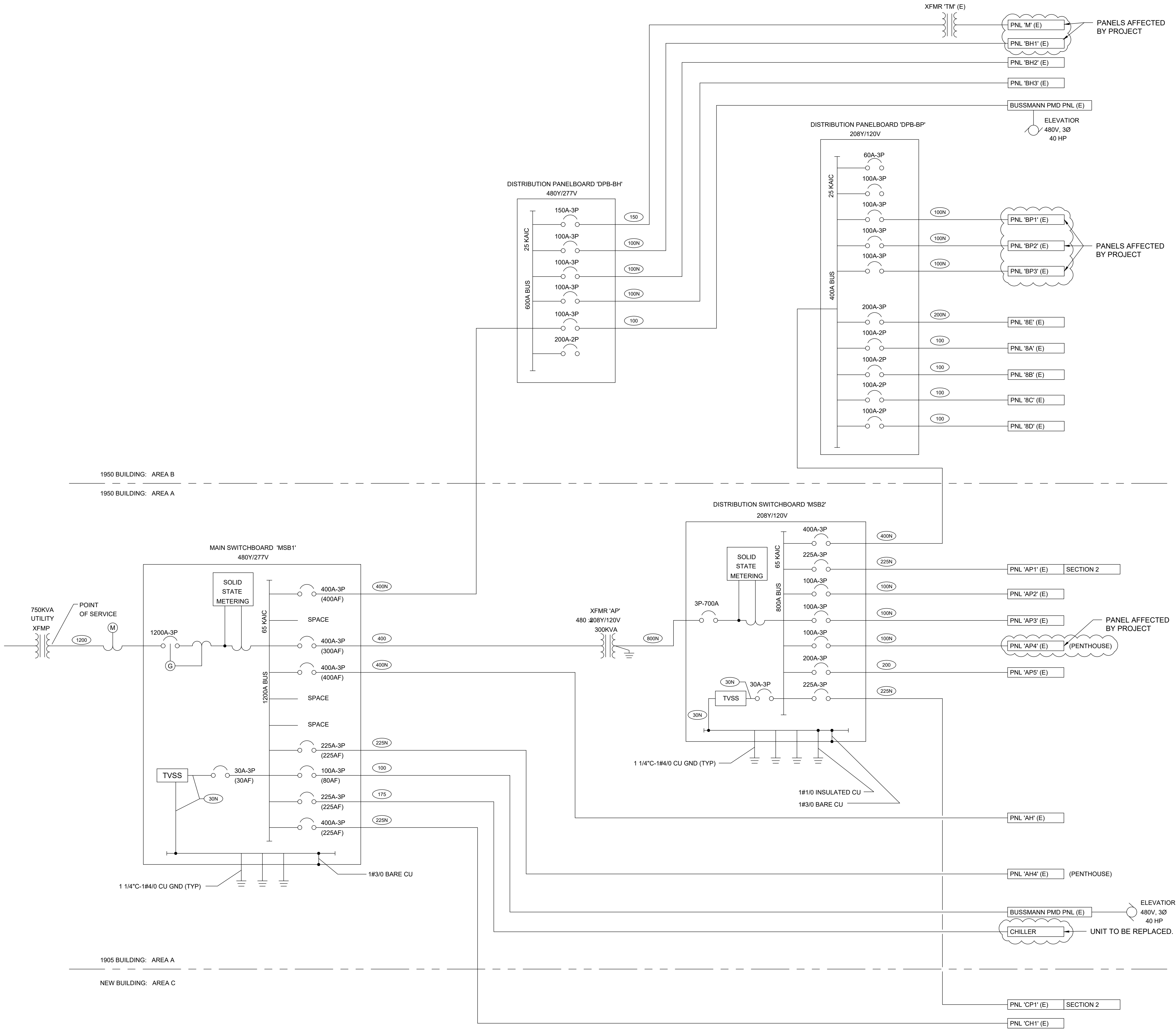
1. VERIFY VOLTAGE, PHASE, FLAM/CA OF EACH CONNECTION WITH EQUIPMENT SUPPLIER PRIOR TO ROUGH-IN. NOTIFY ARCHITECT/ENGINEER WHEN SCHEDULED SUPPLY WILL NOT MEET NEC REQUIREMENTS.
2. OUTLETS, DISCONNECTS, CONTROLLERS, AND EQUIPMENT CONNECTIONS FOR OUTDOOR EQUIPMENT SHALL BE WEATHER PROOF.
3. LOCATION OF OUTLETS, DISCONNECTS, CONTROL DEVICES, AND EQUIPMENT CONNECTIONS ARE TO BE LOCATED IN FIELD BY THE CONTRACTOR. UNLESS OTHERWISE INDICATED ON PLANS, INSTALL SCHEDULED DISCONNECTS AND CONTROL DEVICES INSIGHT OF EQUIPMENT. ARRANGE WIRING AND EQUIPMENT TO AVOID INTERFERENCE WITH OTHER WORK AND TO MAXIMIZE ACCESSIBILITY FOR MAINTENANCE.
4. COORDINATE WITH THE OTHER INSTALLING CONTRACTORS TO ENSURE NEC REQUIRED ACCESS TO DISCONNECTS IS PROVIDED FOR EACH PIECE OF EQUIPMENT.
5. PROVIDE ELECTRICAL CONNECTIONS FOR MECHANICAL FURNISHED SMOKE DETECTORS IN HEATING AND COOLING SYSTEMS PER INTERNATIONAL MECHANICAL CODE. SEE MECHANICAL DIVISION EQUIPMENT SCHEDULES FOR ADDITIONAL UNITS RATED OVER 2000 CFM AND PROVIDE DUCT DETECTOR AS REQUIRED.
6. FIRE ALARM IS EXISTING: FIRELITE ALARMS BY HONEYWELL. MODIFY AND EXTEND SYSTEM AS REQUIRED TO NEW DEVICES. PORTIONS OF EXISTING RACEWAY SYSTEM MAY BE REUSED. TWO COMPLETE PLANS AND SPECIFICATIONS FOR THE FIRE ALARM SYSTEM MODIFICATION SHALL BE SUBMITTED TO AUTHORITY HAVING JURISDICTION FOR REVIEW AND APPROVAL PRIOR TO SYSTEM INSTALLATION.
7. WIRING BETWEEN EQUIPMENT DISCONNECT AND POINT OF CONNECTION SHALL COMPLY WITH NEC BASED ON EQUIPMENT NAMEPLATE RATING EXCEPT MINIMUM BRANCH CIRCUIT RATING
8. SIZE OF DISCONNECT SWITCH AND MOTOR STARTER SHALL BE SIZED TO COMPLY WITH NEC REQUIREMENTS. WHERE INDICATED MOTOR CONTROL IS NOT LOCATED IN SIGHT OF MOTOR AS DEFINED BY NEC, PROVIDE ADDITIONAL DISCONNECTING MEANS TO COMPLY WITH NEC 430.102.
9. WIRING SIZES ARE BASED ON 80 DEGREE C. FOR AMPACITIES 100 AMPERES AND LESS. FOR FEEDERS LESS THAN 100 FEET IN LENGTH, CONDUCTOR SIZES MAY BE SELECTED BASED ON 75 DEGREE C. WHERE EQUIPMENT INSTALLED IS LABELED FOR 75 DEGREE C. WIRING.

MECHANICAL EQUIPMENT CONNECTION SCHEDULE													
NAME	DESCRIPTION	LOCATION	MAXIMUM RATINGS					(CU) FEEDER	CIRCUIT#	DISCONNECT		REMARKS	
			KVA	FLA	MCA	MOCP	VOLT/PH	#12 EACH PHASE + NEUTRAL +GND. UNO.		BY	DESCRIPTION		
CH-1	CHILLER	OUTSIDE	96.44	116.0	145.0	200	480	3	2"C-3#2/0 + #6G	MSB1-25,27,29	●		EXISTING FEEDER TO BE REUSED
HP-10A	INDOOR HEAT PUMP	SHERIFF'S OFFICE	6.69	32.2	40.2	45	208	1	3/4"C-3#6 + #10G	M-17,19	●	60AS/2P	
HP-10B	OUTDOOR HEAT PUMP	OUTSIDE	2.83	13.6	17.0	20	208	1		M-21,23	●	30AS/2P	
HP-20	HEAT PUMP	OUTSIDE	2.33	11.2	14.0	20	208	1		BP1-8,10	●	30AS/2P	INDOOR UNIT FED FROM OUTDOOR UNIT
EH-20	ELECTRIC HEATER	MAINT OFFICE	2.25	10.8	13.5	20	208	1		BP1-7,9	●	TOGGLE	
RF-13	FAN	ATTIC	0.86	7.2	9.0	20	120	1		AP4-29	●	TOGGLE	
REF-1	RELIEF FAN	ATTIC	3.82	10.6	13.3	20	208	3		AP4-19,21,23	●	TOGGLE	
REF-2	RELIEF FAN	ROOF	1.58	7.6	9.5	20	208	1		AP4-25,27	●	TOGGLE	
HP-30	VRF OUTDOOR HEAT PUMP	OUTDOOR	13.30	16.0	20.0	30	480	3	3/4"C-3#10 + #10G	BH1-18/20/22	●	30AS/2P	
BC-1	BRANCH CONTROLLER	COMMISSIONER'S OFFICE	0.17	0.8	1.0	20	208	1		BP2-2,4	●	TOGGLE	
CC-8-1	CEILING CASSETTE	COMMISSIONER'S OFFICE	0.47	2.2	2.8	20	208	1		BP2-2,4	●	TOGGLE	
CC-15-2	CEILING CASSETTE	COMMISSIONER'S OFFICE	0.47	2.2	2.8	20	208	1		BP2-2,4	●	TOGGLE	
CC-15-3	CEILING CASSETTE	COMMISSIONER'S OFFICE	0.47	2.2	2.8	20	208	1		BP2-2,4	●	TOGGLE	
CC-8-4	CEILING CASSETTE	COMMISSIONER'S OFFICE	0.47	2.2	2.8	20	208	1		BP2-2,4	●	TOGGLE	
CC-15-5	CEILING CASSETTE	COMMISSIONER'S OFFICE	0.47	2.2	2.8	20	208	1		BP2-2,4	●	TOGGLE	
CC-15-6	CEILING CASSETTE	COMMISSIONER'S OFFICE	0.47	2.2	2.8	20	208	1		BP2-2,4	●	TOGGLE	
CC-15-7	CEILING CASSETTE	COMMISSIONER'S OFFICE	0.47	2.2	2.8	20	208	1		BP2-2,4	●	TOGGLE	
AH-30-8	AIR HANDLER	JURY ROOM	0.70	3.4	4.2	20	208	1		BP3-18,20	●	TOGGLE	
CC-8-9	CEILING CASSETTE	COMMISSIONER'S OFFICE	0.47	2.2	2.8	20	208	1		BP2-2,4	●	TOGGLE	

EQUIPMENT CONNECTION SCHEDULE NOTES:

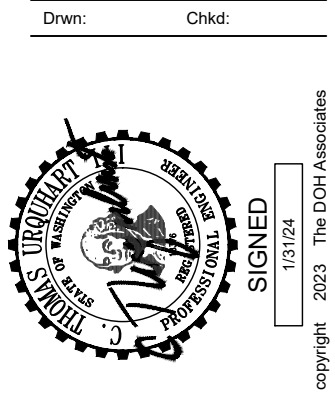
1. VERIFY VOLTAGE, PHASE, FLA/MCA OF EACH CONNECTION WITH EQUIPMENT SUPPLIER PRIOR TO ROUGH-IN. NOTIFY ARCHITECT/ENGINEER WHEN SCHEDULED SUPPLY WILL NOT MEET NEC REQUIREMENTS.
2. OUTLETS, DISCONNECTS, CONTROLLERS, AND EQUIPMENT CONNECTIONS FOR ROOF TOP AND OTHER OUTDOOR EQUIPMENT SHALL BE WEATHER PROOF.
3. LOCATION OF OUTLETS, DISCONNECTS, CONTROL DEVICES, AND EQUIPMENT CONNECTIONS ARE DIAGRAMMATIC AND TO BE LOCATED IN FIELD BY THE CONTRACTOR AS APPROVED BY THE ENGINEER. UNLESS OTHERWISE INDICATED ON PLANS, INSTALL SCHEDULED DISCONNECTS AND CONTROL DEVICES IN SIGHT OF EQUIPMENT. ARRANGE WIRING AND EQUIPMENT TO AVOID INTERFERENCE WITH OTHER WORK AND TO MAXIMIZE ACCESSIBILITY FOR MAINTENANCE AND REPAIRS.
4. COORDINATE WITH THE OTHER INSTALLING CONTRACTORS TO ENSURE NEC REQUIRED ACCESS TO DISCONNECTS IS PROVIDED FOR EACH PIECE OF EQUIPMENT.
5. PROVIDE SMOKE DUCT DETECTORS IN HEATING AND COOLING SYSTEMS PER INTERNATIONAL MECHANICAL CODE. SEE DIVISION 25 EQUIPMENT SCHEDULES FOR ADDITIONAL UNITS RATED OVER 2000 CFM AND PROVIDE DUCT DETECTOR AS REQUIRED.
6. WIRING BETWEEN EQUIPMENT DISCONNECT AND POINT OF CONNECTION SHALL COMPLY WITH NEC BASED ON EQUIPMENT NAMEPLATE RATING EXCEPT MINIMUM BRANCH CIRCUIT RATING SHALL BE 20 AMPERES.
7. SIZE OF DISCONNECT SWITCH AND MOTOR STARTER SHALL BE SIZED TO COMPLY WITH NEC REQUIREMENTS. WHERE INDICATED MOTOR CONTROL IS NOT LOCATED IN SIGHT OF MOTOR AS DEFINED BY NEC, PROVIDE ADDITIONAL DISCONNECTING MEANS TO COMPLY WITH NEC 430.102.
8. WIRING SIZES ARE BASED ON 60 DEGREE C. FOR AMPACITIES 100 AMPERES AND LESS. FOR FEEDERS LESS THAN 100 FEET IN LENGTH, CONDUCTOR SIZES MAY BE SELECTED BASED ON 75 DEGREE C. WHERE EQUIPMENT INSTALLED IS LABELED FOR 75 DEGREE C. WIRING.
9. SCHEDULE LEGEND:
- = FURNISH AND INSTALL NEW UNDER DIVISION 26
 - = INSTALL UNDER DIVISION 26; FURNISHED WITH EQUIPMENT OR BY OTHERS.
 - X = FURNISH AND INSTALL BY OTHERS (NOT DIVISION 26)
 - * = EXISTING, RELOCATED EQUIPMENT

SYMBOL	DESCRIPTION	ABBREV.	DESCRIPTION
	BUBBLE NOTE TAG SYMBOL: # - IDENTIFYING NUMBER	A, AMP	AMPERES
	ELECTRICAL KEYED NOTE TAG SYMBOL: # - IDENTIFYING NUMBER	AC	ALTERNATING CURRENT
AHU-# EF-# XXX-#	SCHEDULED EQUIPMENT CONNECTION (INCLUDE ALL WIRING, DISCONNECTING MEANS, CONTROL AND OTHER REQUIREMENTS SCHEDULED) - SEE MECH	A/C	AIR CONDITIONING
	LUMINAIRE (SIZE/TYPE AS INDICATED)	AIC	AMPERE INTERRUPTING CAPACITY
	EXIT FIXTURE	AL	ALUMINUM
	SINGLE POLE TOGGLE SWITCH	ARCH	ARCHITECTURAL
	DUPLEX RECEPTACLE (NEMA 5-20R)	ATS	AUTOMATIC TRANSFER SWITCH
	GFCI DUPLEX RECEPTACLE (NEMA 5-20R)	AWG	AMERICAN WIRE GAUGE
	JUNCTION BOX	BKR	BREAKER
	MOTOR CONNECTION	C	CONDUIT
	PANELBOARD	C.O.	CONDUIT ONLY
	CIRCUIT BREAKER	CB	CIRCUIT BREAKER
	DISCONNECT SWITCH	CKT	CIRCUIT
	FUSED DISCONNECT SWITCH	CLG	CEILING
	COMBINATION MOTOR STARTER/DISCONNECT	CONC	CONCRETE
	MOTOR STARTER	CT	CURRENT TRANSFORMER
	TRANSFORMER	CU	COPPER
	GENERATOR	DIA	DIAMETER
	AUTOMATIC TRANSFER SWITCH	DIV	DIVISION
	AMP METER	DPDT	DOUBLE POLE, DOUBLE THROW
	VOLT METER	DPST	DOUBLE POLE, SINGLE THROW
	CONTACTOR	EGC	EQUIPMENT GROUND CONDUCTOR
	SMOKE DETECTOR	ELEC	ELECTRIC
		EMT	ELECTRICAL METALLIC TUBING
		EXIST, (E)	EXISTING
		ETR	EXISTING TO REMAIN
		FA	FIRE ALARM
		FC	FOOTCANDLE
		FLA	FULL LOAD AMPS
		GFCI	GROUND FAULT CIRCUIT INTERRUPTER
		GND	GROUND
		HP	HORSEPOWER
		J-BOX	JUNCTION BOX
		KV	KILOVOLTS
		KVA	KILOVOLT AMPERES
		KW	KILOWATTS
		LT	LIGHT
		LTG	LIGHTING
		MAX	MAXIMUM
		MCA	MINIMUM CIRCUIT AMPS
		MCB	MAIN CIRCUIT BREAKER
		MCC	MOTOR CONTROL CENTER
		MCM, KCM	THOUSAND CIRCULAR MILS
		MECH	MECHANICAL
		MIN	MINIMUM
		MLO	MAIN LUGS ONLY
		MOP, MOCP	MAXIMUM OVERCURRENT PROTECTION
		NIC	NOT IN CONTRACT
		NTS	NOT TO SCALE
		OC	ON CENTER
		PB	PULLBOX
		Ø OR PH	PHASE
		PNL	PANEL
		PR	PAIR
		PRI	PRIMARY
		PVC	POLYVINYL CHLORIDE
		RECPT	RECEPTACLE
		REQ	REQUIRED
		RM	ROOM
		SHT	SHEET
		SP	SINGLE POLE
		SPD	SURGE PROTECTIVE DEVICE
		SPDT	SINGLE POLE, DOUBLE THROW
		SPST	SINGLE POLE, SINGLE THROW
		SW	SWITCH
		SWBD	SWITCHBOARD
		TYP	TYPICAL
		UL	UNDERWRITERS LABORATORY
		UF	UNDERFLOOR
		UG	UNDERGROUND
		V	VOLTS
		VA	VOLT AMPERES
		VAC	VOLTS ALTERNATING CURRENT
		VAR	REACTIVE VOLT AMPERES
		W	WATTS
		WP	WEATHERPROOF
		W/	WITH
		W/O	WITHOUT
		XFER	TRANSFER
		XFMR	TRANFORMER
200-4-G	FEEDER CALLOUT X-Y-Z. SEE SCHEDULE.		
200/150-3P	DEVICE SIZE / FUSE OR TRIP RATING - NO. OF POLES		
	HEAVY LINE WEIGHT = NEW WORK (RECEPTACLE SHOWN)		
	STANDARD LINE WEIGHT = EXISTING TO REMAIN (RECEPTACLE SHOWN)		
	BROKEN LINE WORK = ELECTRICAL DEMOLITION (RECEPTACLE SHOWN)		



FEEDER SCHEDULE	
CALL-OUT	DESCRIPTION
30N	3/4"C - 4#10 + #10 GND
100	1 1/2"C - 3#1 + #8 GND
100N	2"C - 4#1 + #8 GND
150	2"C - 3#1/0 + #6 GND
175	2"C - 3#2/0 + #6 GND
200	2"C - 3#3/0 + #6 GND
200N	2 1/2"C - 4#3/0 + #6 GND
225N	2 1/2"C - 4#4/0 + #4 GND
400	4"C - 3#600MCM + #3 GND
400N	4"C - 4#600MCM + #3 GND
800N	(2) 4"C - 4#600MCM + #1/0 GND IN PARALLEL
1200	(4) 3 1/2"C - 4#350 + #3/0 GND IN PARALLEL
COPPER CONDUCTORS UNLESS OTHERWISE NOTED	

HULTZ & BHU
e n g i n e e r s i n c
1111 Fawcett Ave Suite 100
Tacoma, WA 98402
Phone: (253) 383-3257
Fax: (253) 383-3283
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Job Number: 22-223



Douglas County Commissioners
COUNTY COURTHOUSE
Heating, Ventilation, and Air Conditioning Improvements
Waterville, Washington 98858
203 S Rainier Street

The DOH Associates, PS
ARCHITECTS and PLANNERS
7 N Wenatchee Ave Suite 500, Wenatchee, Washington 98801
Telephone (509) 662-4781 Facsimile (509) 663-3253

Job: 2241 Date: 2/15/24
DWG ID: --

ELECTRICAL
ONE-LINE

E0.02

EXISTING

THREE PHASE PANEL SCHEDULE

M

GRID

SECTION: 1 OF 1

LOCATION: SHERIFF OFFICE

VOLTAGE: 208/120 4W

RATING: 100 A

MAIN: BRK

ENCLOSURE

X FLUSH

SURFACE

X NEMA TYPE 1

NEMA TYPE 3R

NEMA TYPE 12

ACCESSORIES

ISOLATED GROUND

SPD

200% NEUTRAL

FEED THRU LUGS

DOUBLE LUGS

SERVICE RATED

A/C ASSEMBLY

SERIES RATED

X 10K

22K

42K

DESCRIPTION	*	VA	BKR	CKT	A	B	C	CKT	BKR	VA	*	DESCRIPTION
LIGHT MAIN OFFICE		390	20/1	1	650	1110		2	20/1	260		LIGHT SW OFFICE
LIGHT MAIN OFFICE		390	20/1	3				4	20/1	720		PLUGS SW OFFICE
LIGHT CENTER OFFICE		260	20/1	5			520	6	20/1	260		LIGHT NW OFFICE
PLUGS CENTER OFFICE		720	20/1	7	1440			8	20/1	720		PLUGS NW OFFICE
PLUGS MAIN OFFICE		900	20/1	9		1260		10	20/1	360		S POWER POLE
EXHAUST FANS		250	20/1	11			610	12	20/1	360		N POWER POLE
MAIN OFFICE A/C		1000	20/1	13	1000			14	20/1			SPARE
CENTER OFFICE A/C		1000	20/1	15		2000		16	20/1	1000		NW OFFICE A/C
220V SW HEAT		500	20/2	17			1500	18	20/1	1000		SW OFFICE A/C
		500	-	19	1000			20	20/2	500		220V MAIN OFFICE HEAT
220V HEAT		500	20/2	21		1000		22	-	500		
		500	-	23			500	24	20/1			SPARE
2ND FLOOR H.W.		1500	20/1	25	1500			26	20/2			SPARE
SPARE			20/2	27		0		28	-			-
		-	-	29			0	30	20/1			SPARE

BREAKER CODE:

A=AFCI, G=GFCI, N=SWITCHED NEUTRAL, S=SHUNT TRIP

K=KEYED, P=PADLOCK ATTACHMENT

R=RED WITH BREAKER LOCK ATTACHMENT

	5590	5370	3130	VA	M
				VA	
				VA	
	5590	5370	3130	VA	SUB-TOTAL

CODE		KVA			KVA	TOTAL LOAD	KVA	AMPS
LIGHTING		1.6	X	125%	2.0	CONNECTED	14.1	39.1
RECEPTACLES		7.8	X	100%	7.8	CALCULATED	14.5	40.2
RECEPTACLES OVER 10K			X	50%				
MOTORS		0.3	X	100%	0.3			
LARGEST MOTOR			X	125%				
KITCHEN			X	100%				
NONCOINCIDENT			X	0%				
REMAINDER		4.5	X	100%	4.5			
EV CHARGER			X	125%				

NOTE *

EXISTING PANEL

LIGHT LINE WEIGHT EQUALS EXISTING

HEAVY LINE WEIGHT EQUALS NEW

REVISED

THREE PHASE PANEL SCHEDULE

M (R)

GRID

SECTION: 1 OF 1

LOCATION: SHERRIFF OFFICE

VOLTAGE: 208/120

4W

RATING: 100 A

MAIN:

BRK

ENCLOSURE

ACCESSORIES

SERVICE RATED

X FLUSH

ISOLATED GROUND

AIC ASSEMBLY

SURFACE

SPD

SERIES RATED

X NEMA TYPE 1

200% NEUTRAL

X 10K

NEMA TYPE 3R

FEED THRU LUGS

22K

NEMA TYPE 12

DOUBLE LUGS

42K

DESCRIPTION	*	VA	BKR	CKT	A	B	C	CKT	BKR	VA	*	DESCRIPTION
LIGHT MAIN OFFICE		390	20/1	1	650	1110		2	20/1	260		LIGHT SW OFFICE
LIGHT MAIN OFFICE		390	20/1	3				4	20/1	720		PLUGS SW OFFICE
LIGHT CENTER OFFICE		260	20/1	5			520	6	20/1	260		LIGHT NW OFFICE
PLUGS CENTER OFFICE		720	20/1	7	1440			8	20/1	720		PLUGS NW OFFICE
PLUGS MAIN OFFICE		900	20/1	9		1260		10	20/1	360		S POWER POLE
EXHAUST FANS		250	20/1	11			610	12	20/1	360		N POWER POLE
SPARE			20/1	13	0			14	20/1			SPARE
SPARE			20/1	15		0		16	20/1			SPARE
HP-10A		3345	45/2*	17			3345	18	20/1			SPARE
		3345	-	19	3345			20	20/2			SPARE
HP-10B		1415	30/2*	21		1415		22	-			
		1415	-	23			1415	24	20/1			SPARE
2ND FLOOR H.W.		1500	20/1	25	1500			26	20/2			SPARE
SPARE			20/2	27		0		28	-			
			-	29			0	30	20/1			SPARE

BREAKER CODE:

A=AFCI, G=GFCI, N=SWITCHED NEUTRAL, S=SHUNT TRIP

K=KEYED, P=PADLOCK ATTACHMENT

R=RED WITH BREAKER LOCK ATTACHMENT

	6935	3785	5890	VA	M (R)
				VA	
				VA	
	6935	3785	5890	VA	SUB-TOTAL

CODE

KVA

KVA

TOTAL LOAD

KVA

AMPS

LIGHTING	1.6	X	125%	2.0		CONNECTED	16.6	46.1
RECEPTACLES	3.8	X	100%	3.8		CALCULATED	17.0	47.2
RECEPTACLES OVER 10K		X	50%					
MOTORS	9.8	X	100%	9.8				
LARGEST MOTOR		X	125%					
KITCHEN		X	100%					
NONCOINCIDENT		X	0%					
REMAINDER	1.5	X	100%	1.5				
EV CHARGER		X	125%					

NOTE *

REPLACE BREAKER WITH NEW

EXISTING PANEL

LIGHT LINE WEIGHT EQUALS EXISTING

HEAVY LINE WEIGHT EQUALS NEW

EXISTING

MSB1

GRID

SECTION: 1 OF 1

LOCATION: SHERIFF OFFICE

VOLTAGE: 480/277

4W

ENCLOSURE

FLUSH

X SURFACE

X NEMA TYPE 1

NEMA TYPE 3R

NEMA TYPE 12

RATING: 1200 A

ACCESSORIES

ISOLATED GROUND

SPD

200% NEUTRAL

FEED THRU LUGS

DOUBLE LUGS

MAIN: BRK

X SERVICE RATED

AIC ASSEMBLY

SERIES RATED

14K

25K

65K

DESCRIPTION	*	VA	BKR	CKT	A	B	C	CKT	BKR	VA	*	DESCRIPTION
(UNLABELED)			80/3	1	0			2				SPACE
				3		0		4				
				5			0	6				
PANEL AH4			225/3	7	0			8	225/3			PANEL CH1
				9		0		10				
SPACE				11			0	12				
				13	0			14	300/3			TRANSFORMER AP (MSB2)
				15		0		16				
PANEL AH			400/3	17			0	18				
				19	0			20	400/3			(UNLABELED) DPB-BH
				21		0		22				
				23			0	24				
CHILLER		47533	225/3	25	47533			26				SPACE
		47533		27		47533		28				
		47533		29			47533	30				
TVSS DISCONNECT			30/3	31	0			32				SPACE
				33		0		34				
				35			0	36				
BREAKER CODE:					47533	47533	47533	VA	MSB1			
A=AFCI, G=GFCI, N=SWITCHED NEUTRAL, S=SHUNT TRIP								VA				
K=KEYED, P=PADLOCK ATTACHMENT								VA				
R=RED WITH BREAKER LOCK ATTACHMENT					47533	47533	47533	VA	SUB-TOTAL			
CODE			KVA	KVA			TOTAL LOAD			KVA	AMPS	
LIGHTING			X	125%			CONNECTED			142.6	171.5	
RECEPTACLES			X	100%			CALCULATED			142.6	171.5	
RECEPTACLES OVER 10K			X	50%								
MOTORS			X	100%								
LARGEST MOTOR			X	125%								
KITCHEN			X	100%								
NONCOINCIDENT			X	0%								
REMAINDER			142.6	X 100%			142.6					
EV CHARGER			X	125%								
NOTE *												
EXISTING PANEL												
LIGHT LINE WEIGHT EQUALS EXISTING												
HEAVY LINE WEIGHT EQUALS NEW												

REVISED

THREE PHASE PANEL SCHEDULE

MSB1 (R)

GRID

SECTION: 1 OF 1

LOCATION: SHERRIFF HOUSE

VOLTAGE: 480/277

4W

RATING: 1200 A

MAIN: BRK

ENCLOSURE

FLUSH

X SURFACE

X NEMA TYPE 1

NEMA TYPE 3R

NEMA TYPE 12

ACCESSORIES

ISOLATED GROUND

SPD

200% NEUTRAL

FEED THRU LUGS

DOUBLE LUGS

X SERVICE RATED

AIC ASSEMBLY

SERIES RATED

14K

25K

65K

DESCRIPTION	*	VA	BKR	CKT	A	B	C	CKT	BKR	VA	*	DESCRIPTION
(UNLABELED)			80/3	1	0			2				SPACE
				3		0		4				
				5			0	6				
PANEL AH4			225/3	7	0			8	225/3			PANEL CH1
				9		0		10				
				11			0	12				
SPACE				13	0			14	300/3			TRANSFORMER AP (MSB2)
				15		0		16				
				17			0	18				
PANEL AH			400/3	19	0			20	400/3			(UNLABELED) DPB-BH
				21		0		22				
				23			0	24				
CHILLER		32147	200/3*	25	32147			26				SPACE
		32147		27		32147		28				
		32147		29			32147	30				
TVSS DISCONNECT			30/3	31	0			32				SPACE
				33		0		34				
				35			0	36				
BREAKER CODE:												
A=AFCI, G=GFCI, N=SWITCHED NEUTRAL, S=SHUNT TRIP												
K=KEYED, P=PADLOCK ATTACHMENT												
R=RED WITH BREAKER LOCK ATTACHMENT												
CODE												
KVA												
KVA												
TOTAL LOAD												
KVA												
AMPS												
CONNECTED												
CALCULATED												
NOTE *												
CHANGE TRIP UNIT IN BREAKER TO 200A.												
EXISTING PANEL												
LIGHT LINE WEIGHT EQUALS EXISTING												
HEAVY LINE WEIGHT EQUALS NEW												

EXISTING

BH1

GRID

SECTION: 1 OF 1

LOCATION: SHERIFF OFFICE

VOLTAGE: 480/277

4W

RATING: 100 A

MAIN: BRK

SERVICE RATED

ACCESSORIES

FLUSH

ISOLATED GROUND

SPD

200% NEUTRAL

FEED THRU LUGS

DOUBLE LUGS

X 14K

25K

65K

SERIES RATED

A/C ASSEMBLY

DESCRIPTION	*	VA	BKR	CKT	A	B	C	CKT	BKR	VA	*	DESCRIPTION
120, 121 BASE HEAT		1500	20/1	1	3000			2	20/1	1500		BASE HEAT - SHERIFF'S
122, 123 BASE HEAT		1500	20/1	3		1500		4	20/1			SPARE
127 BASE HEAT		1000	20/1	5			1000	6	20/1			SPARE
127, 129 BASE HEAT		1500	20/1	7	1500			8	20/1			SPARE
130 BASE HEAT		1000	20/1	9		1000		10	20/1			SPARE
138, 139 BASE HEAT		1500	20/1	11			2500	12	20/1	1000		LIGHTS HEARING ROOM
132, 135 BASE HEAT		1500	20/1	13	2500			14	20/1	1000		LIGHTS HEARING ROOM
131 BASE HEAT		1000	20/1	15		1500		16	20/1	500		OUTSIDE LIGHTS BY HEARING ROOM
FAN COIL HEARING ROOM		2000	30/3	17			2000	18	20/1			SPARE
		2000		19	2000			20	20/1			SPARE
		2000		21		2000		22	20/1			SPARE
SPACE				23			0	24				SPACE
SPACE				25	0			26				SPACE
SPACE				27		0		30				SPACE
SPACE				29			0	28				SPACE

BREAKER CODE:

A=AFCI, G=GFCI, N=SWITCHED NEUTRAL, S=SHUNT TRIP

K=KEYED, P=PADLOCK ATTACHMENT

R=RED WITH BREAKER LOCK ATTACHMENT

	9000	6000	5500	VA	BH1
				VA	
				VA	
	9000	6000	5500	VA	SUB-TOTAL

CODE	KVA			KVA	TOTAL LOAD	KVA	AMPS
LIGHTING	2.5	X	125%	3.1	CONNECTED	20.5	24.7
RECEPTACLES		X	100%		CALCULATED	21.1	25.4
RECEPTACLES OVER 10K		X	50%				
MOTORS		X	100%		NOTE *		
LARGEST MOTOR		X	125%				
KITCHEN		X	100%				
NONCOINCIDENT		X	0%				
REMAINDER	18.0	X	100%	18.0			
EV CHARGER		X	125%				

EXISTING PANEL

LIGHT LINE WEIGHT EQUALS EXISTING

HEAVY LINE WEIGHT EQUALS NEW

REVISED			THREE PHASE PANEL SCHEDULE										
BH1 (R)			VOLTAGE: 480/277			4W		RATING: 100 A			MAIN: BRK		
GRID			ENCLOSURE			ACCESSORIES			SERVICE RATED				
SECTION: 1 OF 1			FLUSH			ISOLATED GROUND			A/C ASSEMBLY				
LOCATION: SHERRIFF OFFICE			X SURFACE			SPD			SERIES RATED				
			X NEMA TYPE 1			200% NEUTRAL			X 14K				
			NEMA TYPE 3R			FEED THRU LUGS			25K				
			NEMA TYPE 12			DOUBLE LUGS			65K				
DESCRIPTION		*	VA	BKR	CKT	A	B	C	CKT	BKR	VA	*	DESCRIPTION
120, 121 BASE HEAT			1500	20/1	1	3000			2	20/1	1500		BASE HEAT - SHERIFF'S
122, 123 BASE HEAT			1500	20/1	3		1500		4	20/1			SPARE
127 BASE HEAT			1000	20/1	5			1000	6	20/1			SPARE
127, 129 BASE HEAT			1500	20/1	7	1500			8	20/1			SPARE
130 BASE HEAT			1000	20/1	9		1000		10	20/1			SPARE
138, 139 BASE HEAT			1500	20/1	11			2500	12	20/1	1000		LIGHTS HEARING ROOM
132, 135 BASE HEAT			1500	20/1	13	2500			14	20/1	1000		LIGHTS HEARING ROOM
131 BASE HEAT			1000	20/1	15		1500		16	20/1	500		OUTSIDE LIGHTS BY HEARING ROOM
FAN COIL HEARING ROOM			2000	30/3	17			6433	18	30/3*	4433		HP-30
			2000		19	6433			20		4433		
			2000		21		6433		22		4433		
SPACE					23			0	24				SPACE
SPACE					25	0		0	26				SPACE
SPACE					27				28				SPACE
SPACE					29			0	30				SPACE
BREAKER CODE:						13433	10433	9933	VA BH1 (R)				
A=AFCI, G=GFCI, N=SWITCHED NEUTRAL, S=SHUNT TRIP									VA				
K=KEYED, P=PADLOCK ATTACHMENT									VA				
R=RED WITH BREAKER LOCK ATTACHMENT						13433	10433	9933	VA SUB-TOTAL				
CODE		KVA		X	125%	KVA			TOTAL LOAD		KVA		AMPS
LIGHTING		2.5		X	125%	3.1			CONNECTED		33.8		40.7
RECEPTACLES				X	100%				CALCULATED		34.4		41.4
RECEPTACLES OVER 10K				X	50%								
MOTORS		13.3		X	100%	13.3			NOTE *				
LARGEST MOTOR				X	125%				* REPLACE BREAKER WITH NEW				
KITCHEN				X	100%								
NONCOINCIDENT				X	0%								
REMAINDER		18.0		X	100%	18.0							
EV CHARGER				X	125%								
EXISTING PANEL LIGHT LINE WEIGHT EQUALS EXISTING HEAVY LINE WEIGHT EQUALS NEW													

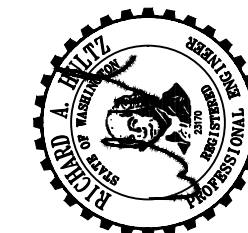
EXISTING THREE PHASE PANEL SCHEDULE															
BP3		VOLTAGE: 208/120		4W		RATING: 100 A		MAIN:		BRK					
GRID		ENCLOSURE		ACCESSORIES		SERVICE RATED		AIC ASSEMBLY		SERIES RATED					
SECTION: 1 OF 1		FLUSH		ISOLATED GROUND											
LOCATION: SHERRIFF OFFICE		X SURFACE		SPD											
		X NEMA TYPE 1		200% NEUTRAL		X 10K									
		NEMA TYPE 3R		FEED THRU LUGS		22K									
		NEMA TYPE 12		DOUBLE LUGS		42K									
DESCRIPTION	*	VA	BKR	CKT	A	B	C	CKT	BKR	VA	*	DESCRIPTION			
BATH		360	15/2	1	1080			2	20/1	720		OFFICE #1 OUTLET			
EXHAUST FAN		500		3		1220		4	20/1	720		OFFICE #2 OUTLET			
UPSTAIRS OUTLET		720	20/1	5			1440	6	20/1	720		OFFICE #3 OUTLET			
COPY MACHINE		1000	20/1	7	2500			8	20/1	1500		REFRIDGERATOR			
SNACK COUNTER		1500	20/1	9		2040		10	20/2	540		WORK STATION #2 OUTLET			
SNACK COUNTER		1500	20/1	11			2040	12		540		WORK STATION #1&2 OUTLET			
RTU PLUG		360	20/1	13	900			14	20/2	540		WORK STATION #1 OUTLET			
ROOF TOP HEAT PUMP		1200	20/2	15		1740		16		540		WORK STATION #1 OUTLET			
SPACE				17			1200	18				SPACE			
SPACE				19	0			20				SPACE			
SPACE				21		0		22				SPACE			
SPACE				23			0	24				SPACE			
SPACE				25	0			26				SPACE			
SPACE				27		0		28				SPACE			
SPACE				29			0	30				SPACE			
BREAKER CODE:				4480		5000	4680	VA BP3							
A=AFCI, G=GFCI, N=SWITCHED NEUTRAL, S=SHUNT TRIP								VA							
K=KEYED, P=PADLOCK ATTACHMENT								VA							
R=RED WITH BREAKER LOCK ATTACHMENT				4480		5000	4680	VA SUB-TOTAL							
CODE		KVA			KVA			TOTAL LOAD	KVA	AMPS					
LIGHTING			X	125%				CONNECTED	14.2	39.3					
RECEPTACLES	10.0	X	100%		10.0			CALCULATED	13.5	37.6					
RECEPTACLES OVER 10K	1.3	X	50%	0.6											
MOTORS	2.9	X	100%	2.9											
LARGEST MOTOR		X	125%												
KITCHEN		X	100%												
NONCOINCIDENT		X	0%												
REMAINDER		X	100%												
EV CHARGER		X	125%												
								EXISTING PANEL							
								LIGHT LINE WEIGHT EQUALS EXISTING							
								HEAVY LINE WEIGHT EQUALS NEW							

EXISTING THREE PHASE PANEL SCHEDULE															
BP2		VOLTAGE: 208/120		4W		RATING: 100 A		MAIN:		BRK					
GRID		ENCLOSURE		ACCESSORIES		SERVICE RATED		AIC ASSEMBLY		SERIES RATED					
SECTION: 1 OF 1		FLUSH		ISOLATED GROUND											
LOCATION: SHERRIFF OFFICE		X SURFACE		SPD											
		X NEMA TYPE 1		200% NEUTRAL		X 10K									
		NEMA TYPE 3R		FEED THRU LUGS		22K									
		NEMA TYPE 12		DOUBLE LUGS		42K									
DESCRIPTION	*	VA	BKR	CKT	A	B	C	CKT	BKR	VA	*	DESCRIPTION			
OUTLETS N. OFFICE		540	20/2	1	540			2	20/1			SPARE			
LIGHTS CONFERENCE		500		3		500		4	20/1			SPARE			
BATH RM LIGHTS		500	20/1	5			500	6	20/1			SPARE			
JURY ROOM A/C		1000	20/2	7	1000			8	20/1			SPARE			
		1000		9		1000		10	20/1			SPARE			
KITCHENETTEOUTLETS WSU EXT.		1500	20/2	11		1500	1500	12	20/1			SPARE			
		1500		13	1500			14				SPARE			
SPACE				15		0		16				SPARE			
SPACE				17			0	18				SPARE			
SPACE				19	0			20				SPARE			
SPACE				21		0		22				SPARE			
SPACE				23			0	24				SPARE			
SPACE				25	0			26				SPARE			
SPACE				27		0		28				SPARE			
SPACE				29			0	30				SPARE			
BREAKER CODE:				3040		1500	2000	VA BP2							
A=AFCI, G=GFCI, N=SWITCHED NEUTRAL, S=SHUNT TRIP								VA							
K=KEYED, P=PADLOCK ATTACHMENT								VA							
R=RED WITH BREAKER LOCK ATTACHMENT				3040		1500	2000	VA SUB-TOTAL							
CODE		KVA			KVA			TOTAL LOAD	KVA	AMPS					
LIGHTING		1.0	X	125%	1.3			CONNECTED	6.5	18.2					
RECEPTACLES	3.5	X	100%		3.5			CALCULATED	6.8	18.8					
RECEPTACLES OVER 10K		X	50%												
MOTORS	2.0	X	100%	2.0											
LARGEST MOTOR		X	125%												
KITCHEN		X	100%												
NONCOINCIDENT		X	0%												
REMAINDER		X	100%												
EV CHARGER		X	125%												
								EXISTING PANEL							
								LIGHT LINE WEIGHT EQUALS EXISTING							
								HEAVY LINE WEIGHT EQUALS NEW							

REVISED										THREE PHASE PANEL SCHEDULE									
BP3 (R)		VOLTAGE: 208/120				4W		RATING: 100 A				MAIN:		BRK					
GRID		<u>ENCLOSURE</u>				<u>ACCESSORIES</u>				<u>SERVICE RATED</u>									
SECTION: 1 OF 1		FLUSH				ISOLATED GROUND				<u>AIC ASSEMBLY</u>									
LOCATION: SHERRIFF OFFICE		X SURFACE				SPD				SERIES RATED									
		X NEMA TYPE 1				200% NEUTRAL				X 10K									
		NEMA TYPE 3R				FEED THRU LUGS				22K									
		NEMA TYPE 12				DOUBLE LUGS				42K									
DESCRIPTION		*	VA	BKR	CKT	A	B	C	CKT	BKR	VA	*	DESCRIPTION						
BATH			360	15/2	1	1080			2	20/1	720		OFFICE #1 OUTLET						
EXHAUST FAN			500		3		1220		4	20/1	720		OFFICE #2 OUTLET						
UPSTAIRS OUTLET			720	20/1	5			1440	6	20/1	720		OFFICE #3 OUTLET						
COPY MACHINE			1000	20/1	7	2500			8	20/1	1500		REFRIDGERATOR						
SNACK COUNTER			1500	20/1	9		2040		10	20/2	540		WORK STATION #2 OUTLET						
SNACK COUNTER			1500	20/1	11			2040	12		540		WORK STATION #1&2 OUTLET						
RTU PLUG			360	20/1	13	900			14	20/2	540		WORK STATION #1 OUTLET						
ROOF TOP HEAT PUMP			1200	20/2	15		1740		16		540		WORK STATION #1 OUTLET						
			1200		17			1200	18	20/2*			AH-30-B						
SPACE					19	0			20										
SPACE					21		0		22				SPACE						
SPACE					23			0	24				SPACE						
SPACE					25	0			26				SPACE						
SPACE					27		0		28				SPACE						
SPACE					29			0	30				SPACE						
BREAKER CODE:						4480	5000	4680	VA		BP3 (R)								
A=AFCI, G=GFCI, N=SWITCHED NEUTRAL, S=SHUNT TRIP									VA										
K=KEYED, P=PADLOCK ATTACHMENT									VA										
R=RED WITH BREAKER LOCK ATTACHMENT						4480	5000	4680	VA		SUB-TOTAL								
CODE		KVA				KVA		TOTAL LOAD		KVA		AMPS							
LIGHTING		X		125%				CONNECTED		14.2		39.3							
RECEPTACLES		10.0		X 100%		10.0		CALCULATED		13.5		37.6							
RECEPTACLES OVER 10K		1.3		X 50%		0.6													
MOTORS		2.9		X 100%		2.9													
LARGEST MOTOR		X		125%				NOTE *											
KITCHEN		X		100%				* INSTALL NEW BREAKER.											
NONCOINCIDENT		X		0%															
REMAINDER		X		100%															
EV CHARGER		X		125%															
								EXISTING PANEL											
								LIGHT LINE WEIGHT EQUALS EXISTING											
								HEAVY LINE WEIGHT EQUALS NEW											

EXISTING														THREE PHASE PANEL SCHEDULE																													
AP4 GRID SECTION: 1 OF 1 LOCATION: SHERRIFF OFFICE														VOLTAGE: 208/120				4W				RATING: 100 A				MAIN: BRK																	
														<u>ENCLOSURE</u>								<u>ACCESSORIES</u>								<u>SERVICE RATED</u>													
														FLUSH								ISOLATED GROUND								<u>AIC ASSEMBLY</u>													
														X SURFACE								SPD								SERIES RATED													
														X NEMA TYPE 1								200% NEUTRAL								10K													
								NEMA TYPE 3R								FEED THRU LUGS								X 22K																			
								NEMA TYPE 12								DOUBLE LUGS								42K																			
DESCRIPTION														*	VA		BKR		CKT		A		B		C		CKT		BKR		VA		*	DESCRIPTION									
(UNKNOWN)															1500				1		1500						2		20/3				SPARE										
CT. ROOM VID, 3RD FLOOR DOORS															1000				3				1000				4																
(UNKNOWN)															1500				5						1500		6																
(UNKNOWN)															1500				7		1500						8		20/3				SPARE										
CUPOLA															1500				9				1500		10																		
(UNKNOWN)															1500				11						1500		12																
SPARE																			13		0						14		20/3				SPARE										
(UNKNOWN)															1500				15				1500		16																		
CUPLOA															1500				17						1500		18																
SPARE																	20/3		19		0						20		20/3				SPARE										
																			21				0				22																
																			23						0				24														
SPACE																			25		0						26						SPACE										
SPACE																			27				0				28						SPACE										
SPACE																			29						0				30				SPACE										
BREAKER CODE:																		3000		4000		4500				VA		AP4															
A=AFCI, G=GFCI, N=SWITCHED NEUTRAL, S=SHUNT TRIP																										VA																	
K=KEYED, P=PADLOCK ATTACHMENT																												VA															
R=RED WITH BREAKER LOCK ATTACHMENT																		3000		4000		4500						VA		SUB-TOTAL													
CODE														KVA								KVA								TOTAL LOAD				KVA		AMPS							
LIGHTING														X				125%												CONNECTED				11.5		31.9							
RECEPTACLES														X				100%												CALCULATED				11.5		31.9							
RECEPTACLES OVER 10K														X				50%																									
MOTORS														X				100%																									
LARGEST MOTOR														X				125%																									
KITCHEN														X				100%																									
NONCOINCIDENT														X				0%																									
REMAINDER														11.5				X				100%				11.5																	
EV CHARGER																		X				125%																					

REVISED														THREE PHASE PANEL SCHEDULE																		
AP4 (R) GRID SECTION: 1 OF 1 LOCATION: SHERRIFF OFFICE														VOLTAGE: 208/120				4W		RATING: 100 A				MAIN: BRK								
														ENCLOSURE				ACCESSORIES				SERVICE RATED										
														FLUSH				ISOLATED GROUND				AIC ASSEMBLY										
														X SURFACE				SPD				SERIES RATED										
														X NEMA TYPE 1				200% NEUTRAL				10K										
				NEMA TYPE 3R				FEED THRU LUGS				X 22K																				
				NEMA TYPE 12				DOUBLE LUGS				42K																				
DESCRIPTION														*	VA	BKR	CKT	A	B	C	CKT	BKR	VA	*	DESCRIPTION							
(UNKNOWN)															1500		1	1500				2	20/3				SPARE					
CT. ROOM VID. 3RD FLOOR DOORS															1000		3		1000			4										
(UNKNOWN)															1500		5			1500	6											
(UNKNOWN)															1500		7	1500			8	20/3					SPARE					
CUPOLA															1500		9		1500		10											
(UNKNOWN)															1500		11			1500	12											
SPARE															1500		13	0			14	20/3						SPARE				
(UNKNOWN)															1500		15		1500		16											
CUPLOA															1500		17			1500	18											
REF-1															1273	20/3**	19	1273			20	20/3							SPARE			
															1273		21		1273		22											
															1273		23			1273	24											
REF-2															790	20/2*	25	790		790	26								SPACE			
															790		27				28								SPACE			
RF-13															860	20/1*	29			860	30								SPACE			
BREAKER CODE:																			5063	6063	6633	VA	AP4 (R)									
A=AFCI, G=GFCI, N=SWITCHED NEUTRAL, S=SHUNT TRIP																						VA										
K=KEYED, P=PADLOCK ATTACHMENT																						VA										
R=RED WITH BREAKER LOCK ATTACHMENT																			5063	6063	6633	VA	SUB-TOTAL									
CODE														KVA				KVA				TOTAL LOAD				KVA		AMPS				
LIGHTING																X	125%								CONNECTED	17.8	49.3					
RECEPTACLES																X	100%								CALCULATED	17.8	49.3					
RECEPTACLES OVER 10K																X	50%															
MOTORS																X	100%															
LARGEST MOTOR																X	125%															
KITCHEN																X	100%															
NONCOINCIDENT																X	0%															
REMAINDER															17.8	X	100%	17.8														
EV CHARGER																X	125%															



1/31/2024



HULTZ & BROS.
e n g i n e e r s i n c

1111 Fawcett Ave Suite 100 Tacoma, WA 98402
Phone: (253) 383-3257 Fax: (253) 383-3283
general@hultzbu.com Job Number: 22-223

Douglas County Commissioners
COUNTY COURTHOUSE
Heating, Ventilation, and Air Conditioning Improvements
203 S Rainier Street
Waterville, Washington 98858

The DOH Associates, PS
ARCHITECTS and PLANNERS

Job: 2241 Date: 2/15/24

GENERAL NOTES:

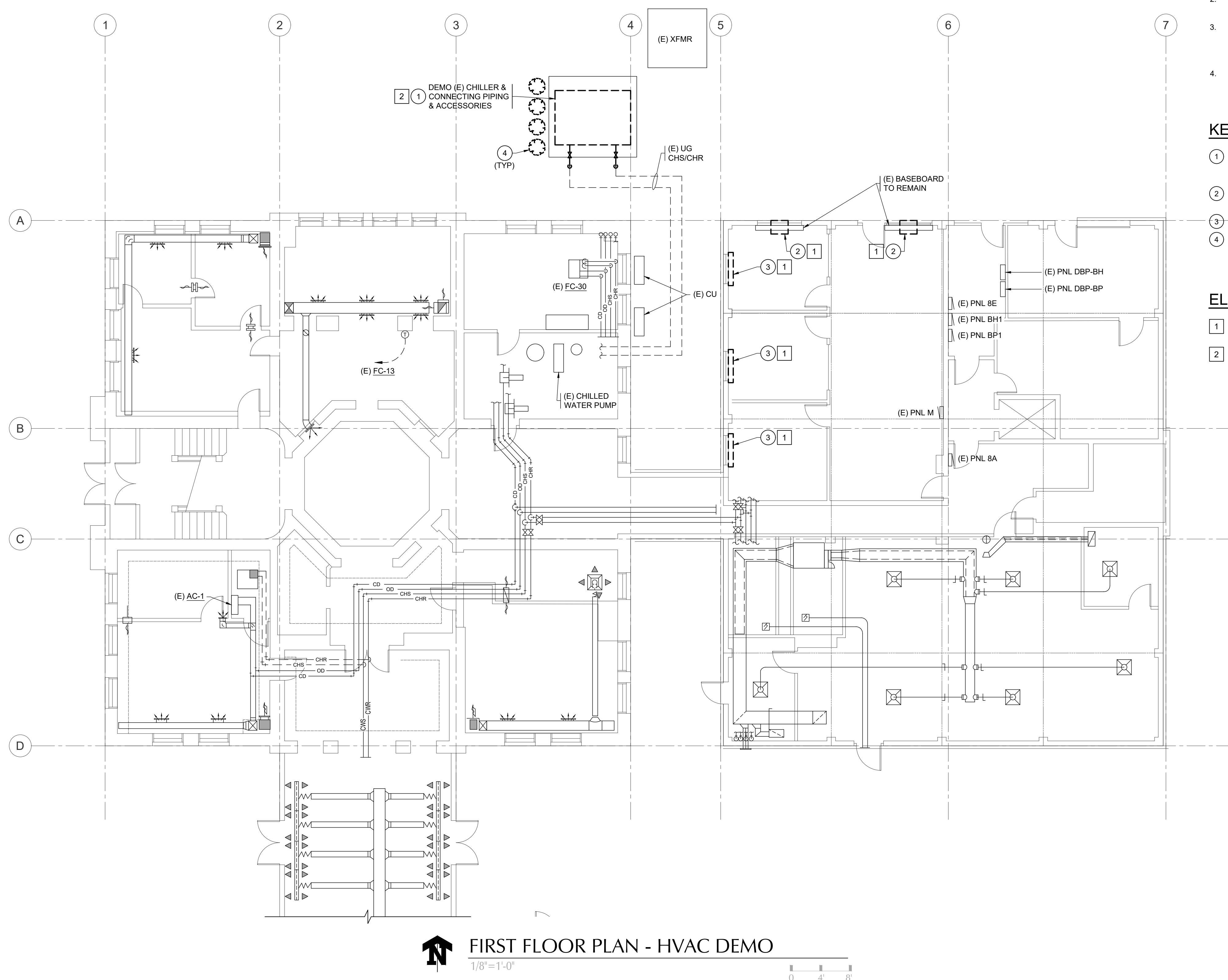
1. SEE DEMOLITION NOTES & MECHANICAL GENERAL NOTES SHEET M0.1.
2. EXISTING MECHANICAL ITEM LOCATIONS ARE APPROXIMATE AND REQUIRE FIELD VERIFICATION.
3. EXISTING ITEMS SHOWN DASHED REPRESENT MAJOR ITEMS TO BE REMOVED (NOT ALL ITEMS SHOWN). SEE GENERAL NOTES, DRAWING NOTES & KEYED NOTES WHICH COVER OTHER MISCELLANEOUS ITEMS TO BE REMOVED.
4. REMOVE (E) CEILING AS REQUIRED TO ACCOMPLISH WORK. REINSTALL TO MATCH (OR BE BETTER THAN) EXISTING CONDITIONS.

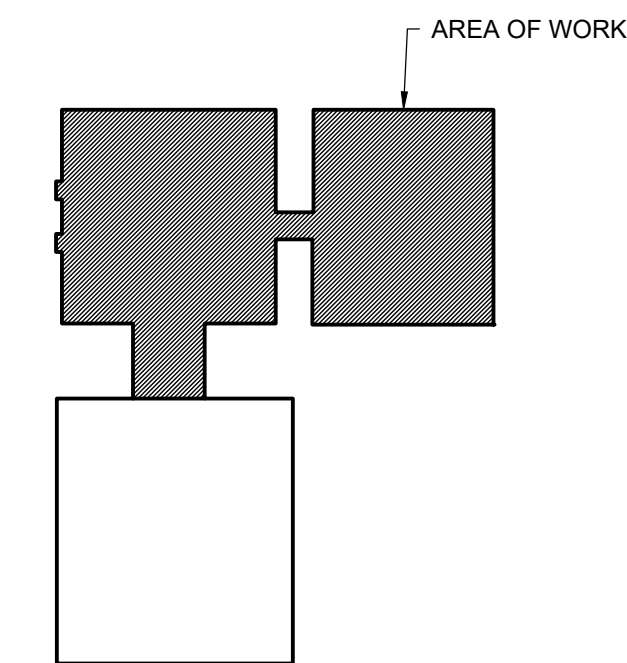
KEYED NOTES:

- ① EXISTING CHILLER DATA:
CARRIER 303TR070-4020RS, 171.6 MCA, 225 MOP, 460V/1PH W/
115V/1PH 30A CONTROL CKT.
- ② DEMO (E) THROUGH WALL AC UNIT (PTAC), PROVIDE INFILL OF
WALL OPENING (SEE ARCH DWGS).
- ③ DEMO (E) ELECTRIC BASEBOARD HEATER.
- ④ REMOVE (E) PLANTS TO ALLOW FOR LARGER UNIT &
CONCRETE PAD.

ELECTRICAL KEYED NOTES:

- 1 REMOVE CONDUIT AND WIRING COMPLETE BACK TO SOURCE PANEL.
- 2 DISCONNECT EXISTING WIRING TO ALLOW UNIT TO BE REPLACED. DISCONNECT SWITCH TO BE SALVAGED FOR USE WITH NEW EQUIPMENT.

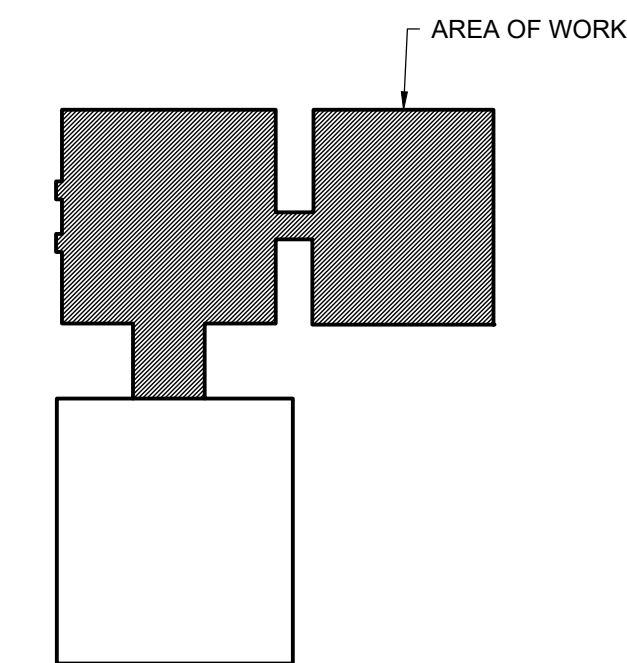




NTS

A horizontal number line with tick marks at 0, 4, and 8. The tick mark at 4 is labeled with a superscripted 'i'.

- 1 REMOVE CONDUIT AND WIRING COMPLETE BACK TO SOURCE PANEL.**



THIRD FLOOR PLAN - HVAC DEMO

ME1.03

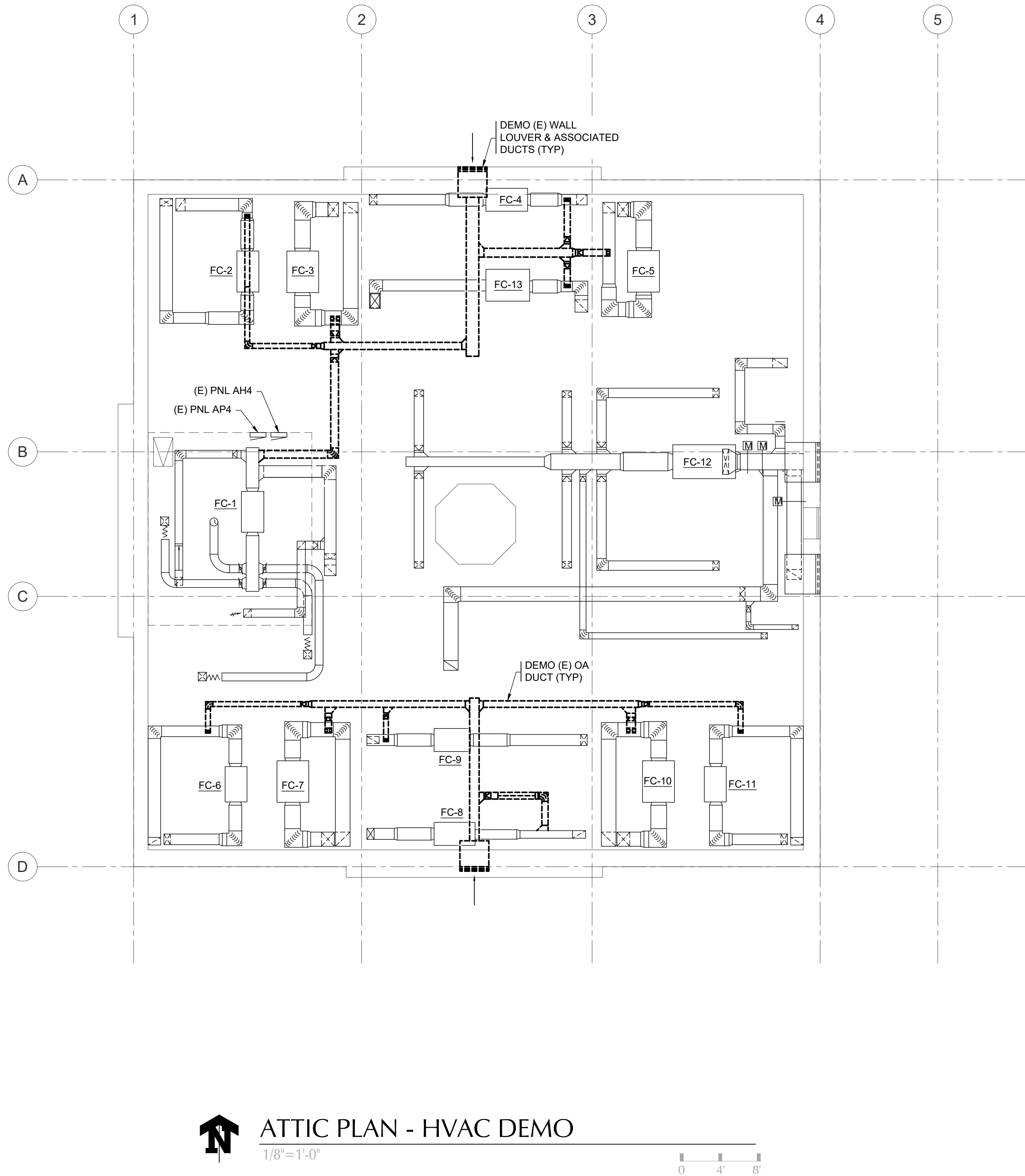
1 REMOVE CONDUIT AND WIRING COMPLETE BACK TO SOURCE PANEL.

- 1 DEMO (E) ELECTRIC BASEBOARD HEATERS.
- 2 DEMO (E) THROUGH WALL AC UNIT (PTAC), PROVIDE INFILL OF WALL OPENING (SEE ARCH DWGS).

- ① DEMO (E) ELECTRIC BASEBOARD HEATERS.

- 2 DEMO (E) THROUGH WALL AC UNIT (PTAC), PROVIDE INFILL OF WALL OPENING (SEE ARCH DWGS).

$$\overline{1/8'' = 1' - 0'}$$

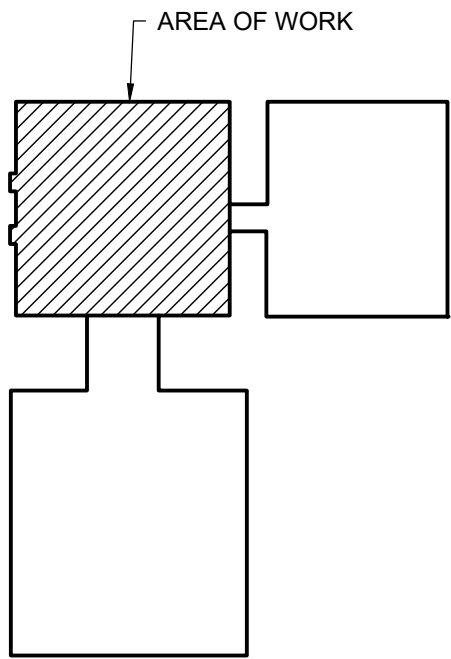
ATTIC PLAN - HVAC DEMO

1/8"=1'-0"

0 4' 8'

GENERAL NOTES:

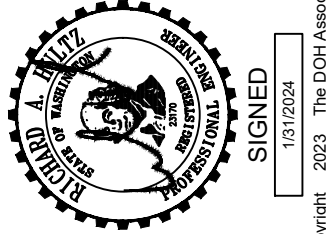
1. SEE DEMOLITION NOTES & MECHANICAL GENERAL NOTES SHEET M0.1.
2. EXISTING MECHANICAL ITEM LOCATIONS ARE APPROXIMATE AND REQUIRE FIELD VERIFICATION.
3. EXISTING ITEMS SHOWN DASHED REPRESENT MAJOR ITEMS TO BE REMOVED (NOT ALL ITEMS SHOWN). SEE GENERAL NOTES, DRAWING NOTES & KEYED NOTES WHICH COVER OTHER MISCELLANEOUS ITEMS TO BE REMOVED.
4. REMOVE (E) CEILING AS REQUIRED TO ACCOMPLISH WORK. REINSTALL TO MATCH (OR BE BETTER THAN) EXISTING CONDITIONS.



KEY PLAN

NTS

HULTZ + BHU
e n g i n e e r s i n c .
1111 Fawcett Ave Suite 100 Tacoma, WA 98402
Phone: (253) 383-3257 Fax: (253) 383-3283
general@hultzbhu.com Job Number: 22-223



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The DOH Associates, PS
ARCHITECTS and PLANNERS
7 N Wenatchee Ave Suite 500, Wenatchee, Washington 98801
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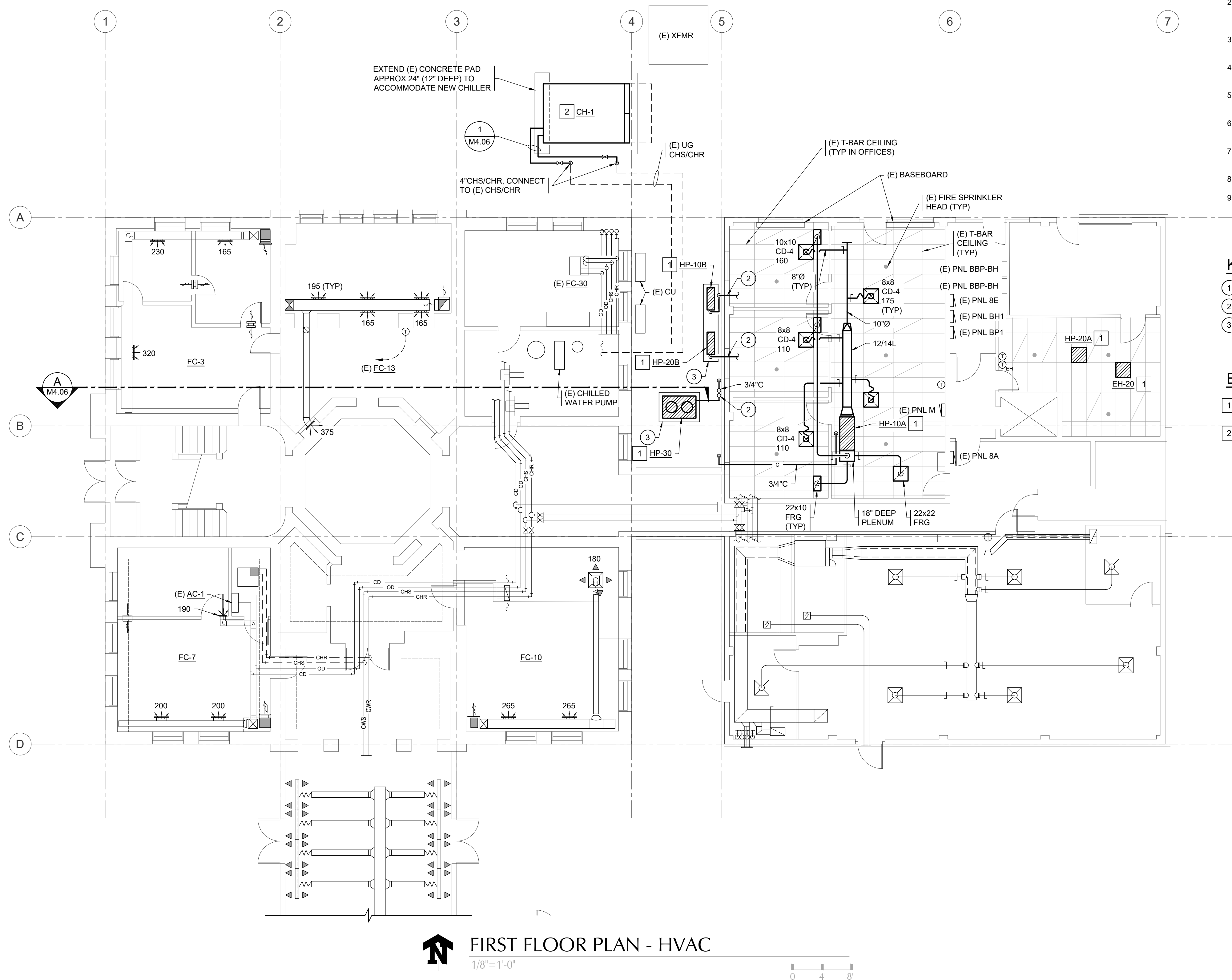


Job: 2241 Date: 2/15/24

DWG ID: --

ATTIC PLAN -
HVAC DEMO

ME1.04



FIRST FLOOR PLAN - HVAC

GENERAL NOTES:

1. SEE MECHANICAL GENERAL NOTES SHEET M0.01.
2. LOCATIONS OF AIR INLETS/OUTLETS ARE APPROXIMATE. FIELD VERIFY LOCATIONS TO SUIT (E) LIGHTS & OTHER BLDG FEATURES.
3. PROVIDE P-TRAP AT EQUIPMENT CONDENSATE CONNECTION WITH 3/4" CONDENSATE DRAIN TO OUTSIDE LANDSCAPE AREA.
4. PROVIDE VIBRATION ISOLATION FOR ALL SUSPENDED EQUIPMENT W/ ROTATING PARTS.
5. VERIFY SPACE AVAILABLE FOR MECHANICAL ITEMS PRIOR TO ORDERING MATERIALS; NO AS-BUILT DRAWINGS EXIST.
6. EXISTING CEILING IS T-BAR TYPE (UNO). REMOVE/REINSTALL AS REQUIRED TO ACCOMPLISH WORK.
7. GRID LINES ARE FOR REFERENCE ONLY & DO NOT RELATE TO STRUCTURAL ELEMENTS.
8. SEE REFRIGERANT PIPING DIAGRAM SHEET M4.09.
9. REMOVE (E) CEILING & PROVIDE CUTTING/ PATCHING AS REQUIRED TO PERFORM WORK. REINSTALL CEILING & RESTORE ALL AREAS TO PRE-CONSTRUCTION CONDITIONS OR BETTER.

KEYED NOTES:

- 1 CEILING SPACE USED AS A RETURN AIR PLENUM.
- 2 RG/RL TO INDOOR UNIT.
- 3 PROVIDE CONCRETE PAD MIN 6" THICK, MIN 6" LARGER ALL AROUND THAN UNIT.

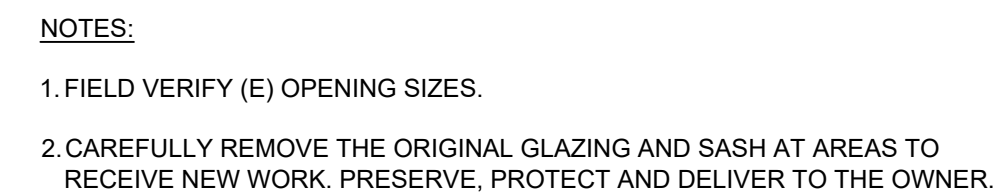
ELECTRICAL KEYED NOTES:

- 1 PROVIDE NEW CONDUIT, WIRING, AND DISCONNECT FOR UNIT. SEE MECHANICAL EQUIPMENT CONNECTION SCHEDULE ON SHEET E0.1.
- 2 MODIFY AND EXTEND EXISTING CONDUIT AND WIRING TO NEW CHILLER. REUSE SALVAGED DISCONNECT SWITCH FOR NEW UNIT. SEE MECHANICAL EQUIPMENT CONNECTION SCHEDULE ON SHEET E0.1 AND ONE-LINE DIAGRAM ON SHEET E0.2.

1 PROVIDE NEW CONDUIT, WIRING, AND DISCONNECT FOR UNIT.
PROVIDE CONNECTION TO VFD & FROM VFD TO FAN. SEE
MECHANICAL EQUIPMENT CONNECTION SCHEDULE ON SHEET E0.1.

1. SEE MECHANICAL GENERAL NOTES SHEET M0.01.
2. LOCATIONS OF EXISTING DUCTWORK & EQUIPMENT ARE APPROXIMATE. FIELD VERIFY LOCATIONS.
3. VERIFY SPACE AVAILABLE FOR MECHANICAL ITEMS PRIOR TO ORDERING OR FABRICATING MATERIALS. AS-BUILT DRAWINGS ARE LIMITED.
4. GRID LINES ARE FOR REFERENCE ONLY & DO NOT RELATE TO STRUCTURAL ELEMENTS.
5. VERIFY SIZES OF ALL EXISTING DUCTS BEING CONNECTED TO OR REVISED. SIZES SHOWN ARE BASED ON OWNER'S AS-BUILTS AND HAVE NOT BEEN FIELD VERIFIED.
6. EXISTING HYDRONIC PIPING & FIRE PIPING MAY REQUIRE REVISIONS. SEE DIVISION 00 & 01 FOR ALLOWANCES TO INCLUDED IN BID.

- ① WRAP FAN INLET DUCT & OUTLET DUCT W/ ACOUSTIC WRAP
WRAP FROM LOUVER TO OUTLET OF SOUND ATTENUATOR.
- ② SOUND ATTENUATOR, SEE DUCT ACCESSORIES
SPECIFICATION SECTION.

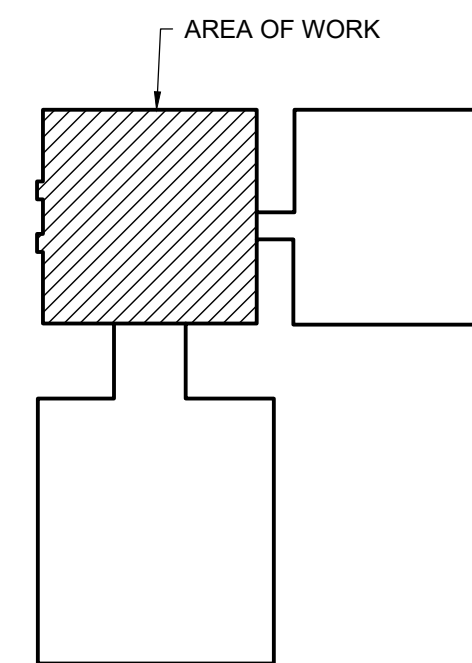


EAST LOUVER DETAIL

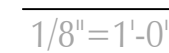
KEY PLAN

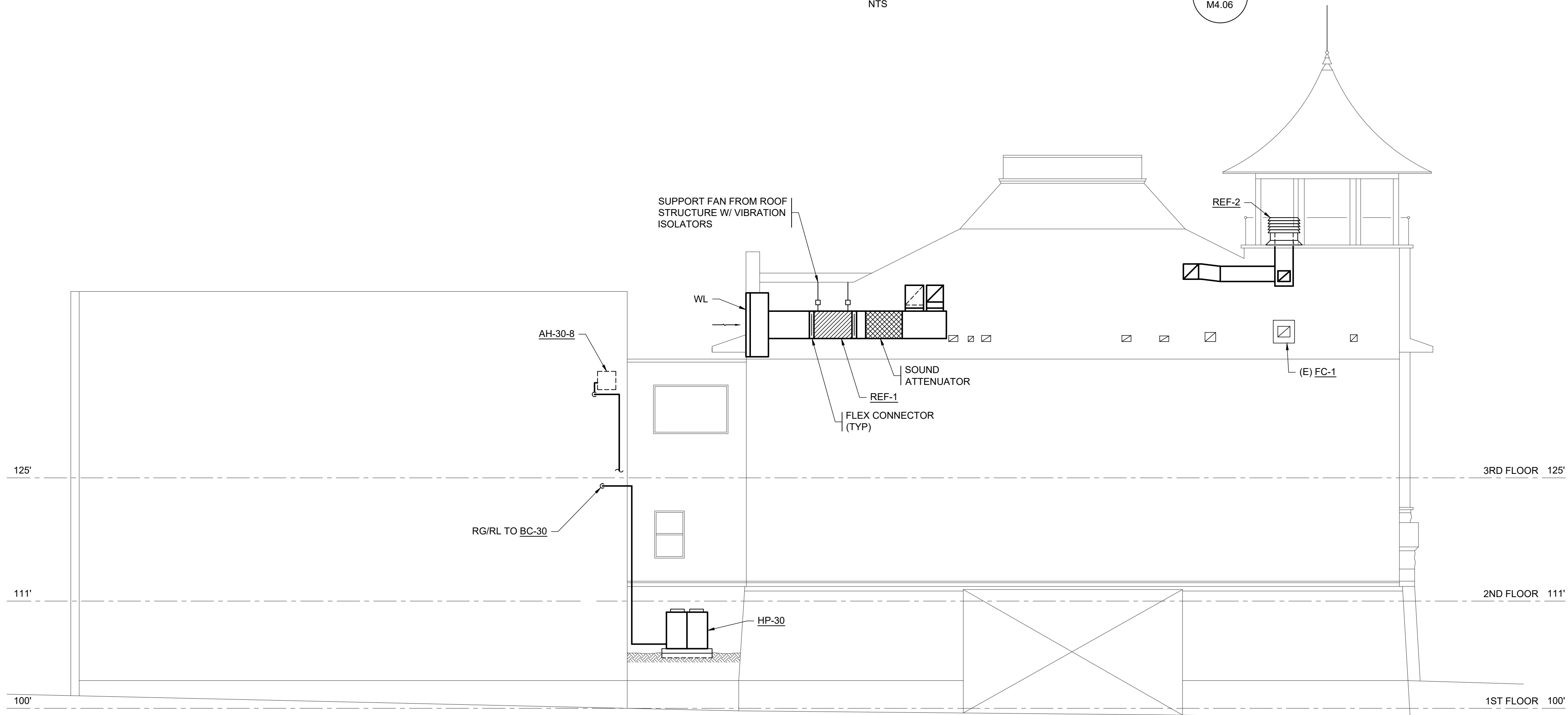
1. SEE MECHANICAL GENERAL NOTES SHEET M0.01.
2. LOCATIONS OF EXISTING DUCTWORK & EQUIPMENT ARE APPROXIMATE. FIELD VERIFY LOCATIONS.
3. VERIFY SPACE AVAILABLE FOR MECHANICAL ITEMS PRIOR TO ORDERING OR FABRICATING MATERIALS. AS-BUILT DRAWINGS ARE LIMITED.
4. GRID LINES ARE FOR REFERENCE ONLY & DO NOT RELATE TO STRUCTURAL ELEMENTS.
5. VERIFY SIZES OF ALL EXISTING DUCTS BEING CONNECTED TO OR REVISED. SIZES SHOWN ARE BASED ON OWNER'S AS-BUILTS AND HAVE NOT BEEN FIELD VERIFIED.

1 PROVIDE NEW CONDUIT, WIRING, AND DISCONNECT FOR UNIT. PROVIDE CONNECTION TO VFD & FROM VFD TO FAN. SEE MECHANICAL EQUIPMENT CONNECTION SCHEDULE ON SHEET E0.1.



NTS

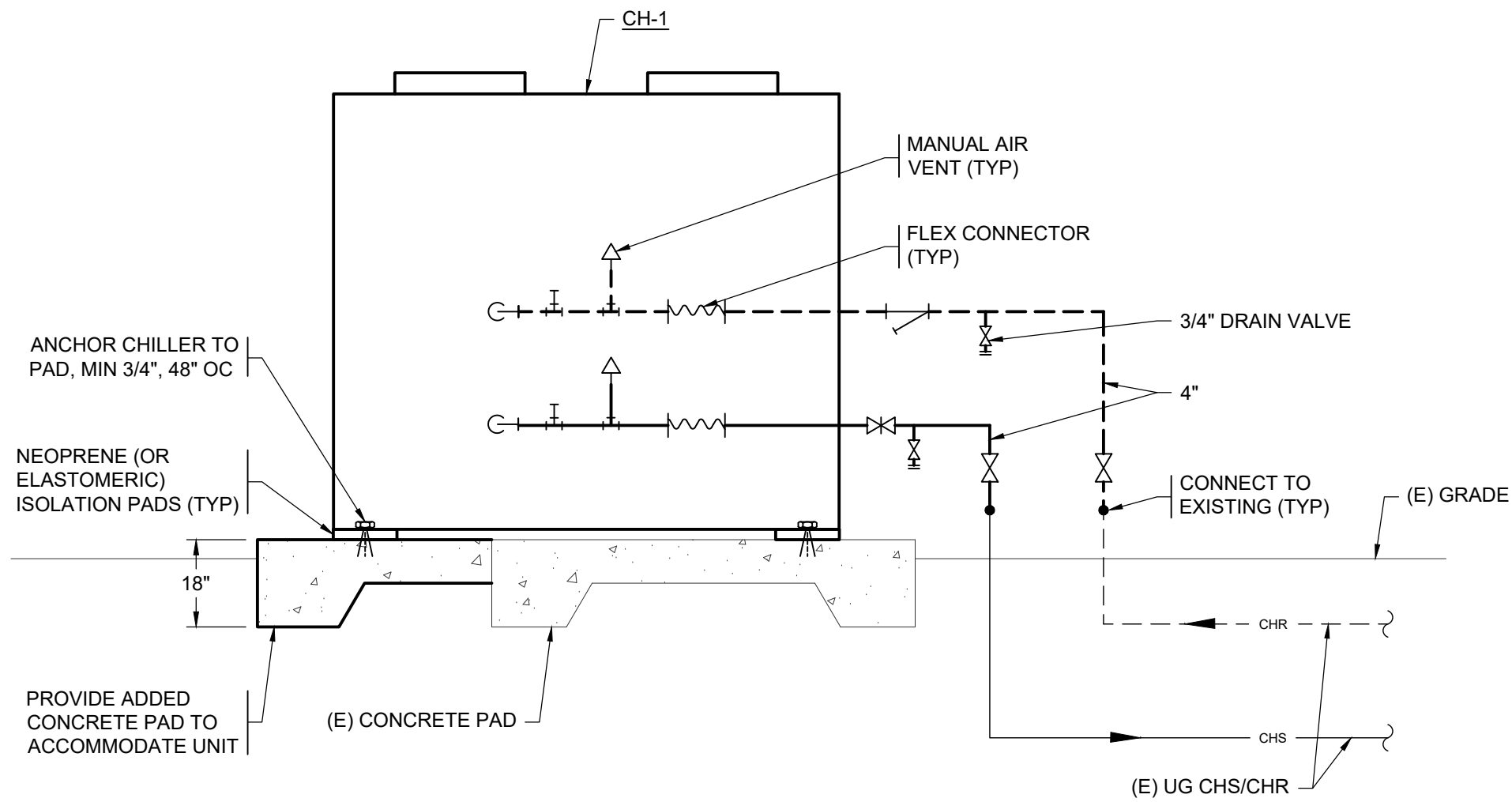




NOTE: NOT ALL WORK SHOWN IN SECTION, SEE FLOOR PLANS.

SECTION
SCALE: 1/8" = 1'-0"

A
M4.06



- NOTES:
1. INSULATE ALL SURFACES & ITEMS IN CONTACT WITH CHILLED WATER SO NO CONDENSATION CAN OCCUR; PROVIDE W/ METAL JACKETING. PROVIDE REMOVABLE INSULATION AT STRAINER, BALANCING VALVE, & MAV'S.
 2. PROVIDE WATER TREATMENT & SYSTEM GLYCOL AS SPECIFIED.

CHILLER CONNECTIONS

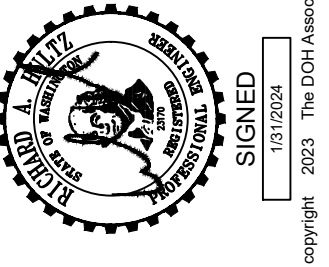
NTS

1
M4.06

MECHANICAL
DETAILS

M4.06

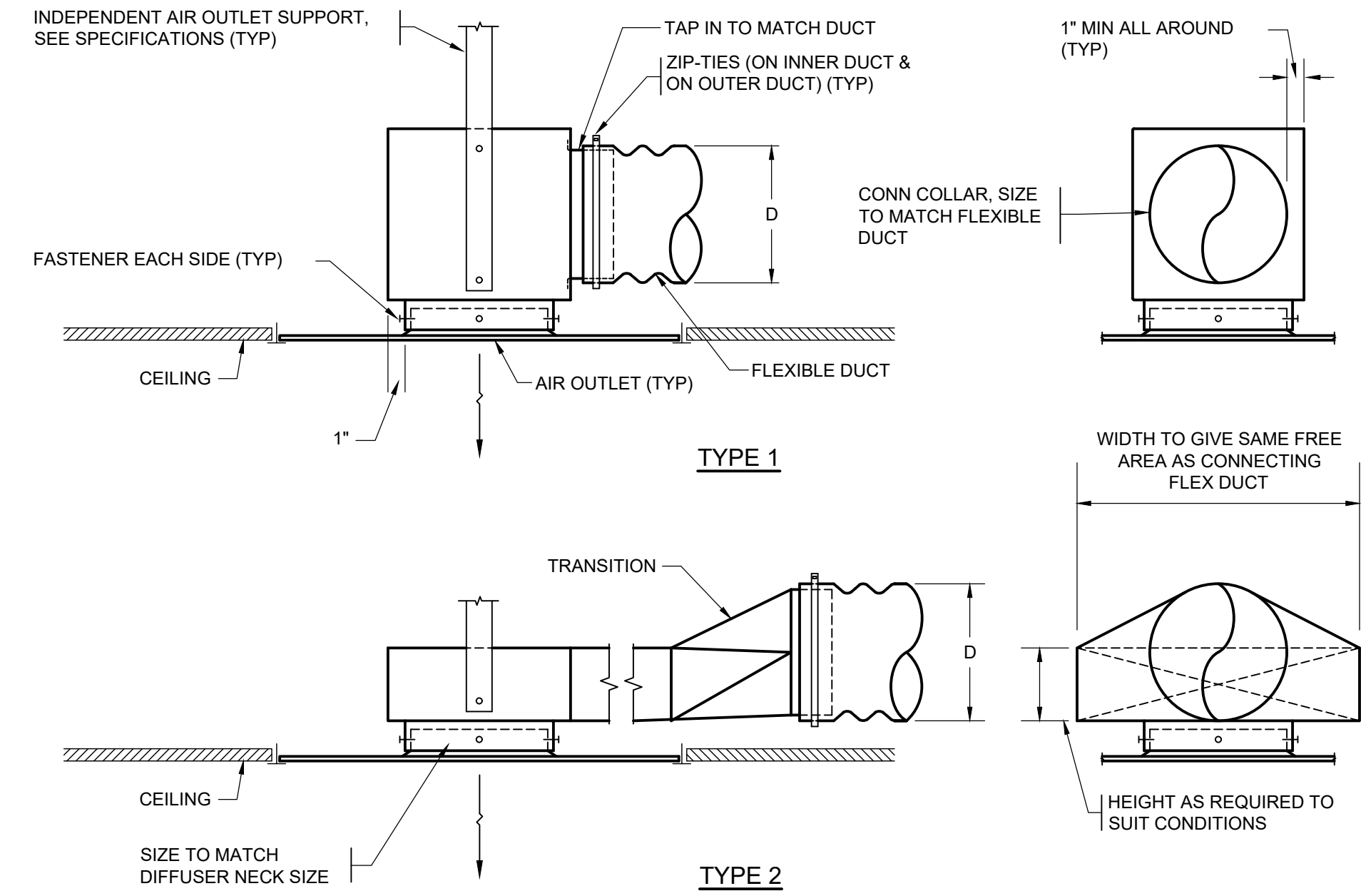
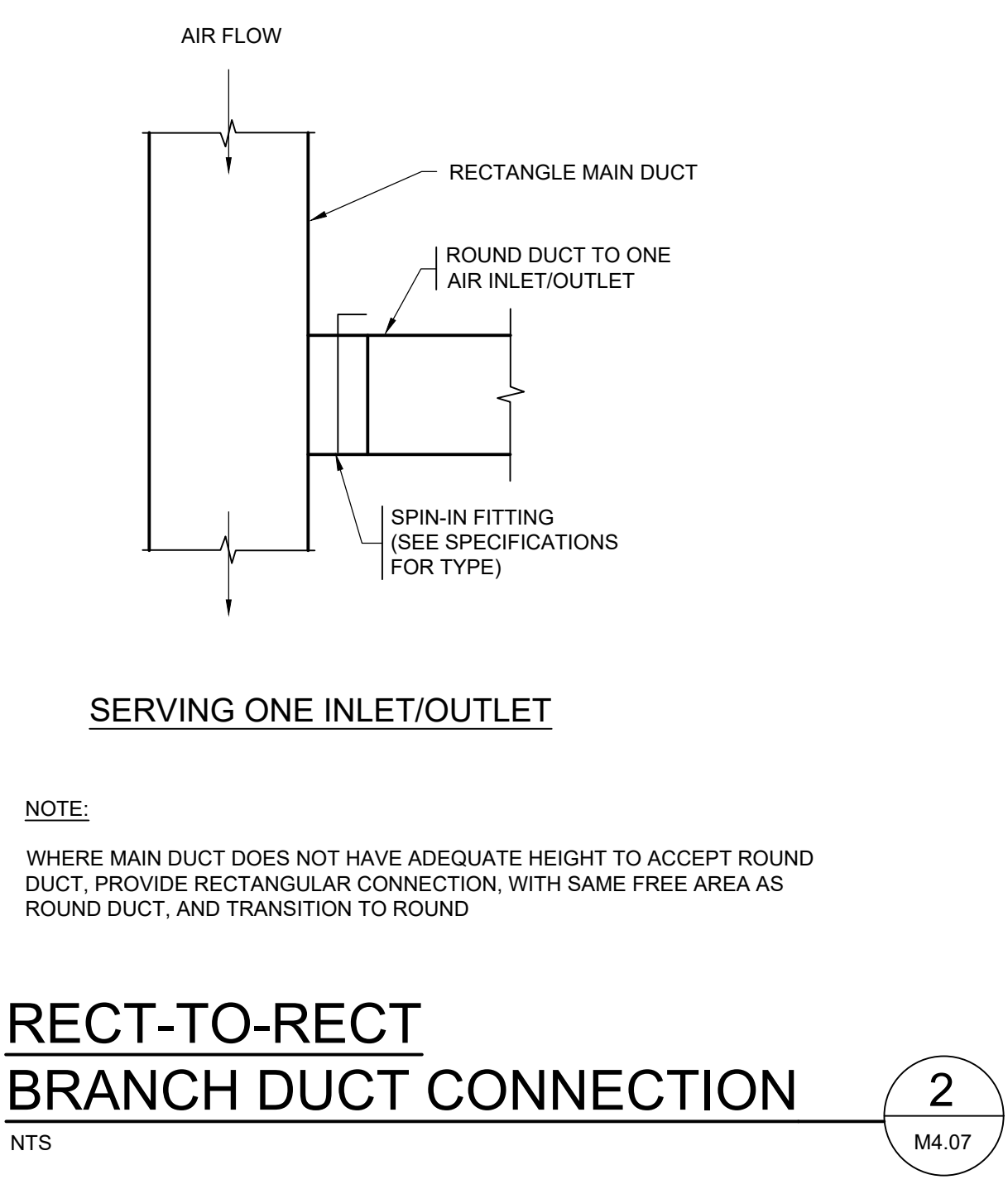
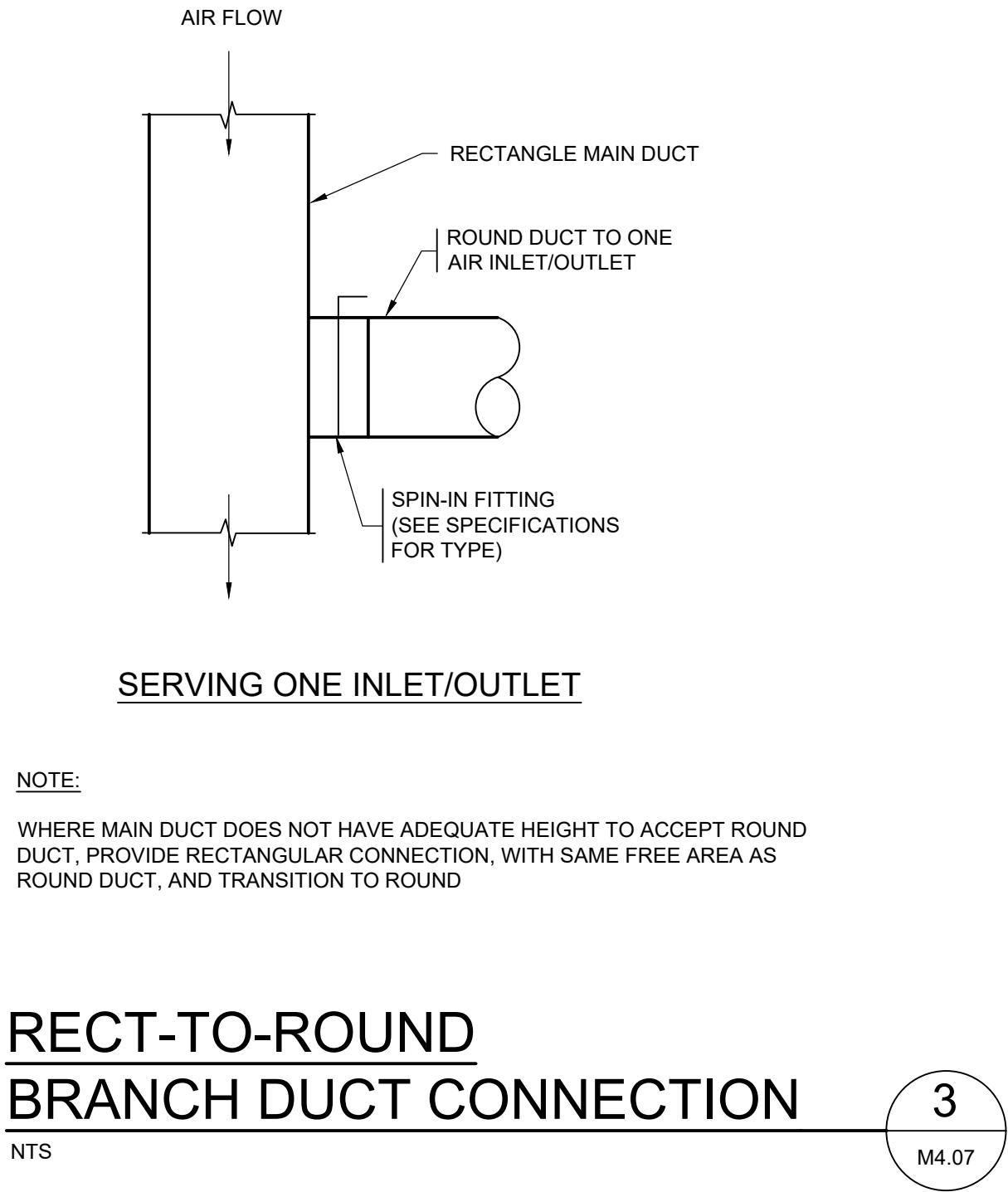
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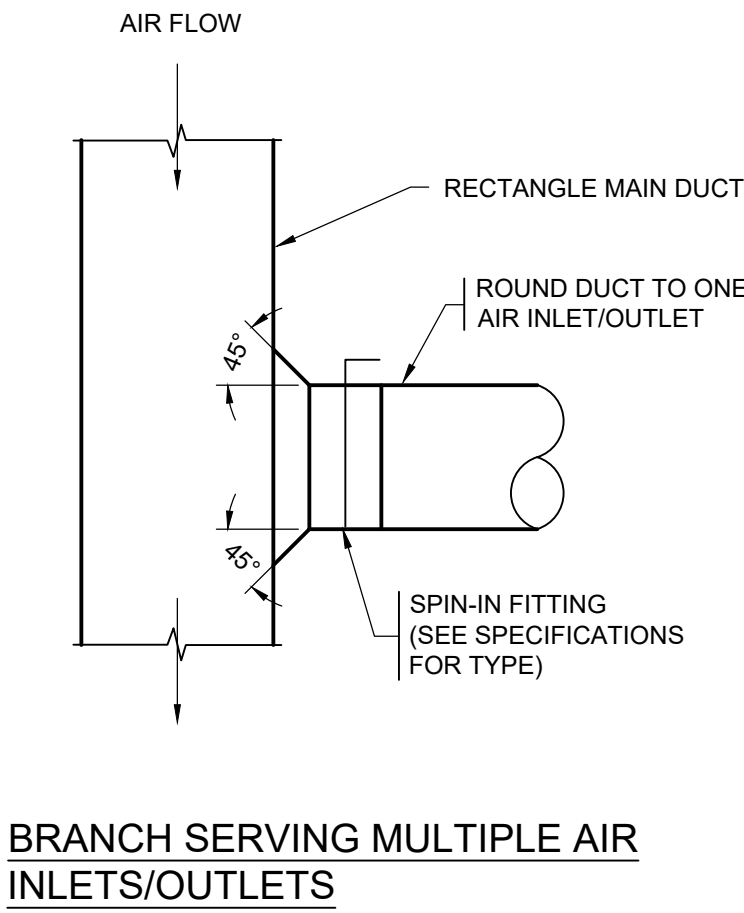
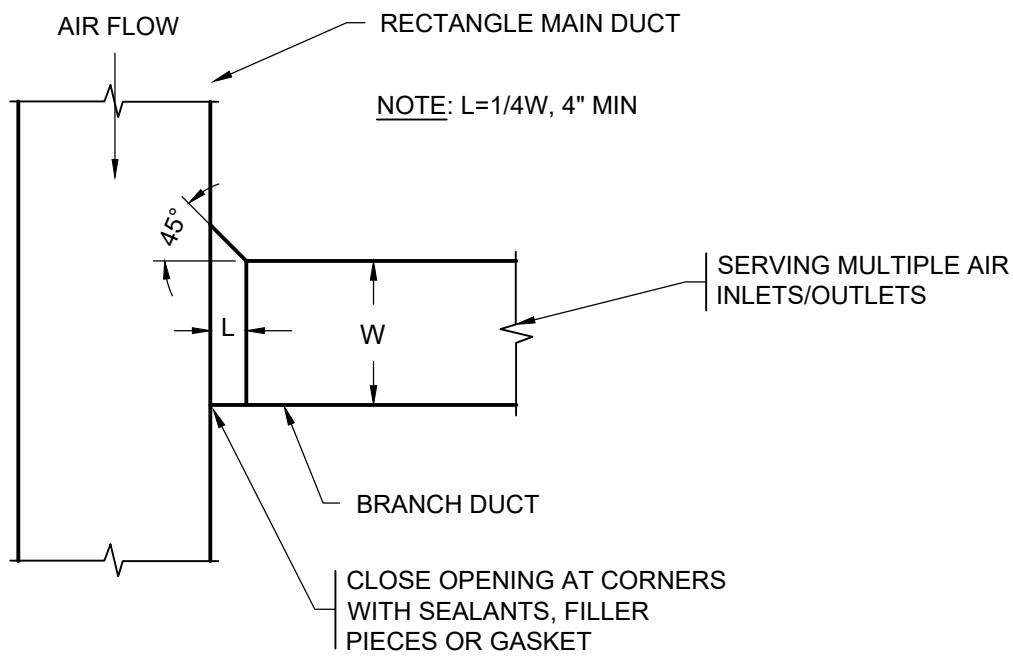
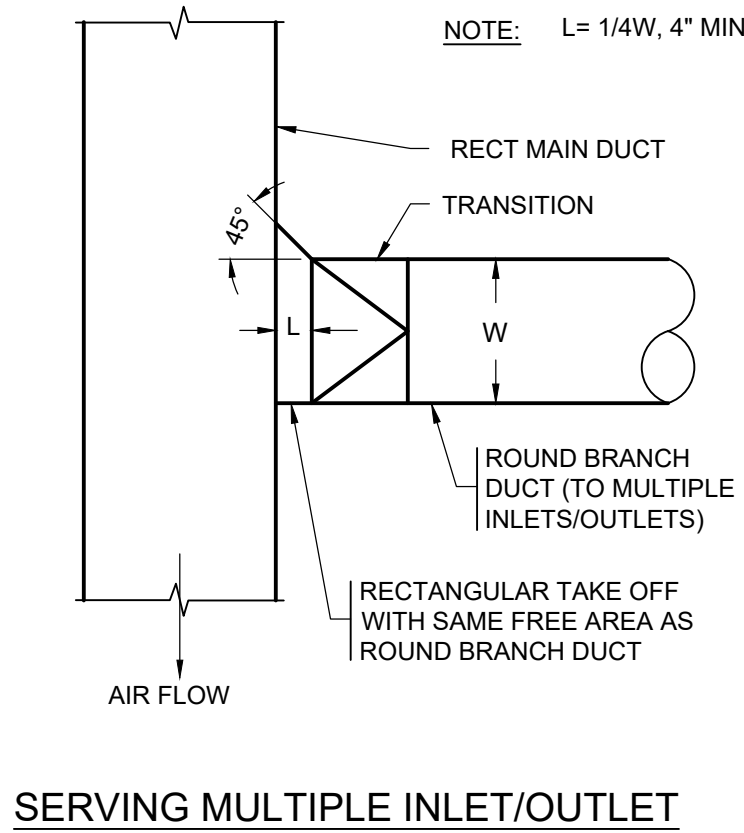
Job: 2241 Date: 2/15/24
DWG ID: -



- NOTES:
1. PROVIDE TYPE 1 UNLESS BUILDING CONDITIONS REQUIRE LOWER PROFILE - THEN USE TYPE 2.
 2. CONSTRUCT PLENUM BOXES OF MIN. 26 GA. GALV. STEEL.
 3. NOT ALLOWED AT TRANSFER DUCTS.
 4. LINE PLENUM & DUCT TO AIR OUTLET WHERE SO NOTED ON PLANS OR IN SPECIFICATIONS.
 5. CEILING TYPE & AIR OUTLET FRAME STYLE MAY VARY FROM THAT DEPICTED.
 6. NOT ALL SUPPORTS ARE SHOWN FOR CLARITY.
 7. SEE SPECIFICATIONS FOR ADDITIONAL REQUIREMENTS.

TIGHT CONDITION - FLEX DUCT AIR OUTLET CONNECTION

1
M4.07
NTS



NOTE:
WHERE MAIN DUCT DOES NOT HAVE ADEQUATE HEIGHT TO ACCEPT ROUND DUCT, PROVIDE RECTANGULAR CONNECTION, WITH SAME FREE AREA AS ROUND DUCT, AND TRANSITION TO ROUND

RECT-TO-ROUND BRANCH DUCT CONNECTION

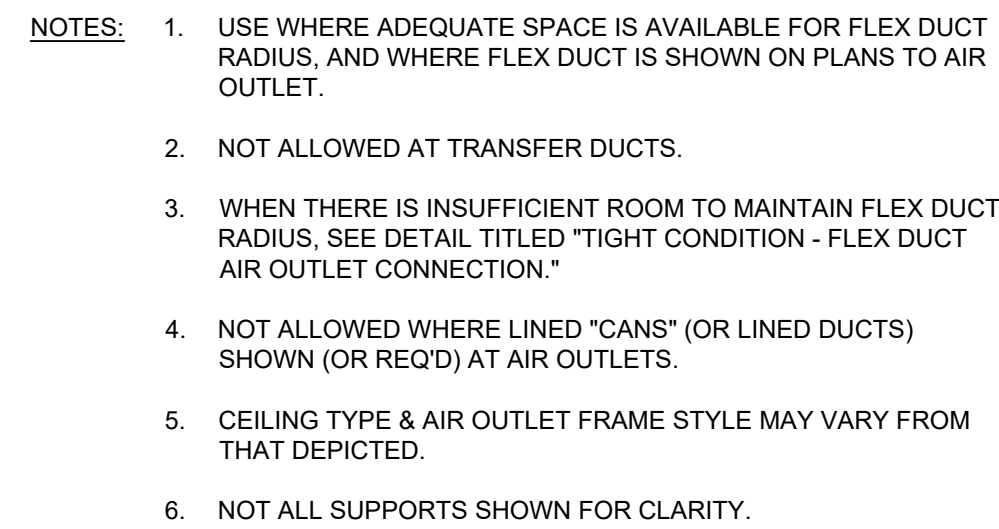
4
M4.07
NTS

RECT-TO-ROUND BRANCH DUCT CONNECTION

6
M4.07
NTS

RECT-TO-RECT BRANCH DUCT CONNECTION

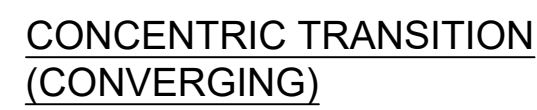
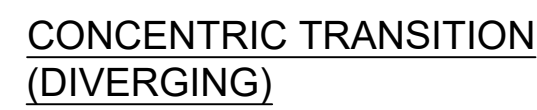
5
M4.07
NTS



NTS

9

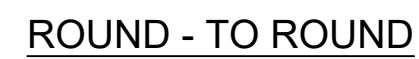
M4.08



NTS

8

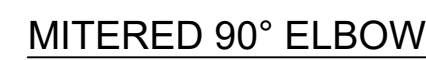
M4.08



NTS

7

M4.08



DIMENSION:	
DIAMETER (INCH)	NUMBER OF VANES
3-9	2
10-14	3
15-19	4
20-60	5
OVER 60	12" MAX. SPACING



NTS

11

M4.08



<u>DIMENSION:</u>	
ANGLE	NUMBER OF GORES
0-35°	2
36-71°	3
72-90°	5

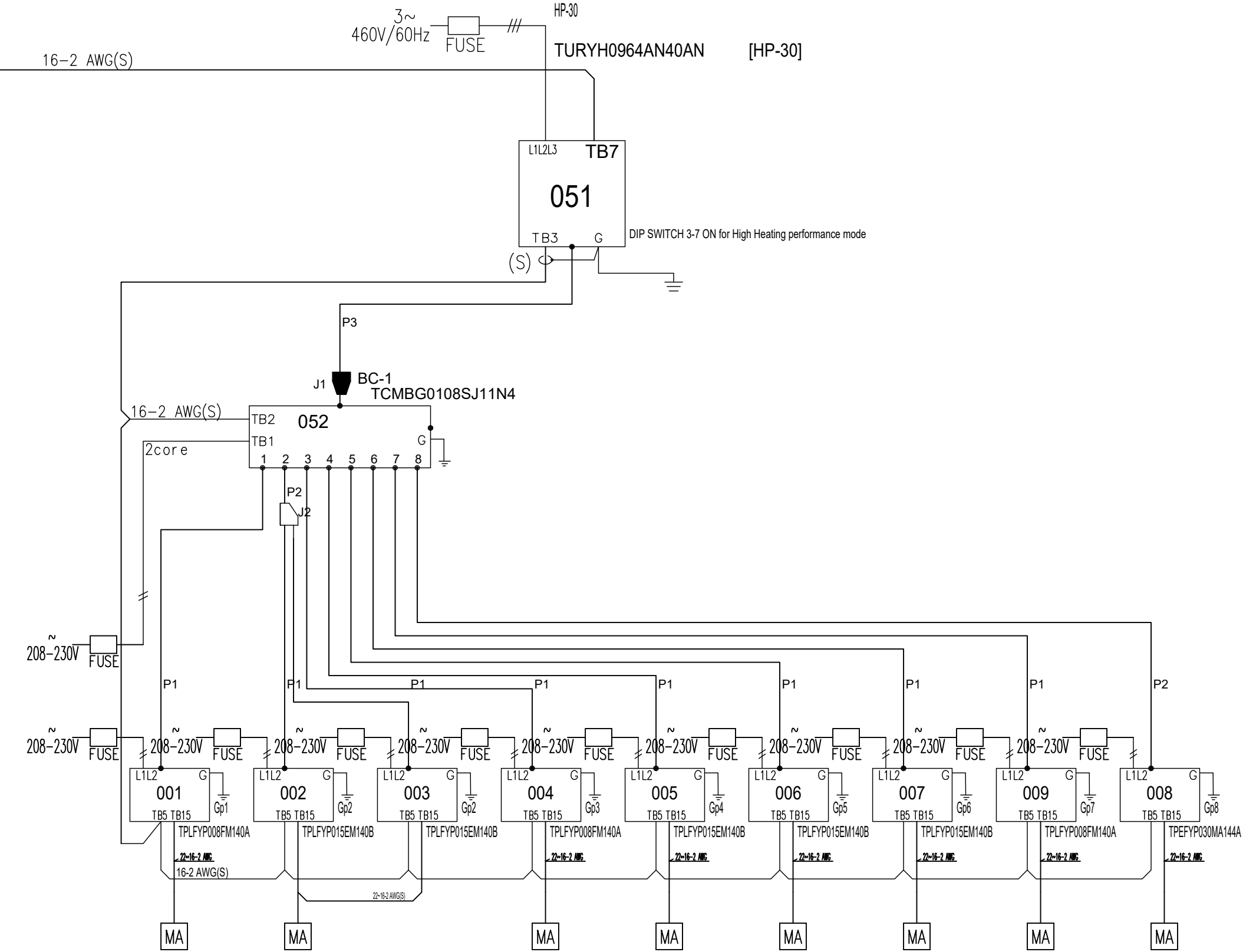
 $R=1.5W$

NTS

10

M4.08

Douglas County Courthouse		
DIAGRAM	SYMBOL	LEGEND
DISPLAY	DESCRIPTION	
---	POWER WIRE	
---	CONTROL WIRE	
---	REF. PIPE	



CC-8-1, CC-15-2, CC-15-3, CC-8-4, CC-15-5, CC-15-6, CC-15-7, CC-8-8, AH-30-8

REMARKS
Originator: Michael Woods
Comments:

REFRIGERANT PIPING DIAGRAM

NTS

1
M4.09

NOTE:
THIS DIAGRAM HAS BEEN PROVIDED BY THE VRF MFR AND IS PRELIMINARY & APPROXIMATE ONLY. IT DOES NOT SHOW ALL REQ'D CONTROL DEVICES NOR ALL CONNECTIONS, NOR A DESIGN FOR OTHER MFR'S. OBTAIN FROM VRF MFR UPDATED DIAGRAM WITH ALL PIPING SIZES, CONTROLS, WIRING, AND RELATED REQUIREMENTS. SUBMIT SHOP DRAWINGS AS SPECIFIED.